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KETCH: A Knowledge-Enhanced Transformer-Based Approach to Suicidal Ideation Detection from Social Media Content

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Abstract. Suicidal ideation (SI), as a psychiatric emergency, requires immediate assistance and intervention. Most people with SI do not actively seek help from mental health professionals, which may result in irreversible consequences. Research has shown that individuals experiencing SI increasingly express their thoughts and emotions on social media platforms, making the latter a viable venue for suicidal ideation detection (SID). This paper proposes, develops, and evaluates a knowledge-enhanced transformer-based approach (KETCH) to SID from social media content. KETCH comprises several key novel design artifacts, including a social media-oriented SI lexicon, a model-level method for integrating domain knowledge (i.e., lexicon) into a state-of-the-art transformer, and aligned dynamic embedding and lexicon-based enhancement that integrate domain relevance and contextual importance of terms to effective SID. We evaluate KETCH's performance with social media data in two different languages collected from distinct platforms, and further examine its generalizability to user-level models for suicide risk prediction and depression detection. The results demonstrate the superior effectiveness, robustness, and generalizability of KETCH via a series of empirical evaluation and a field study. Our research makes multifold research contributions and opens up practical opportunities for timely detection and proactive intervention of SI, which can have far-reaching impacts on public health, the economy, and society.

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Keywords: suicidal ideation detection • social media • domain knowledge • lexicon • transformer • representation enhancement

1. Introduction

Raimundo, a 27-year-old mental health advocate, has had suicidal thoughts since she was a teenager. Talking to people about her struggles in real life hasn't always been her best option. A teacher she confided in once suggested she was "crazy," and she was too ashamed to tell her parents . . . That's why sharing her feelings—positive and negative—with her Facebook friends is important to her. When Raimundo, who had anxiety, depression, and a binge-eating disorder, did post suicidal thoughts on Facebook, nothing happened. A friend messaged her the following day to let her know that they had reported her post and to ask how Facebook had intervened. The company did not reply. Raimundo recalled the gist of her since-deleted post: "I feel like I failed you guys because I'm not feeling well, and it feels like a roller coaster and I'm just not sure it's worth it anymore." "I was really sad because I posted this heartfelt confession of really feeling suicidal and taking my own life, and nobody [from Facebook] talked to me," In her

high school, she lost four friends to suicide.¹ (Extracted from *The Ringer*, April 24, 2017).

This story reflects a glaring problem: suicide is not rare. According to the World Health Organization (WHO), suicide is a global phenomenon. More than 700,000 people worldwide die from suicide annually.² Suicide is the fourth leading cause of death among people between 15 and 29 years of age globally. The National Alliance on Mental Illness reports that 4.8% of adults and 11.8% of young adults 18–25 years of age in the United States have had serious thoughts about committing suicide.³ Suicide has become a major public mental health concern (Yates et al. 2017). Recognizing its significance, the U.S. government activated the nationwide phone number 988 as the Suicide and Crisis Lifeline in July 2022.

Suicidality consists of four sequential stages: suicidal ideation (SI), suicide plan, suicide attempt, and completed suicide (Svetitic and De Leo 2012). Existing literature defines SI as thoughts, considerations, or contemplation of engaging in suicide-related behaviors (Wang et al. 2019, Zortea et al. 2020); suicide plan is a systematic formulation of actions that can lead to self-injury and death (Silverman et al. 2007); and suicide attempt is defined as nonfatal, self-directed, and potentially injurious behaviors with an intent to die (Klonsky et al. 2016). SI is a risk factor for subsequent stages. It can be linked to, yet different from, self-harm and depression. Self-harm is the act of deliberately causing harm or injuries to oneself. It is an expression of personal distress rather than a mental illness. Although SI has been found in 28%–41% of self-injury cases, upward of 85% of self-injury incidents occur with the primary intention of releasing tension rather than of ending life (Yates 2004). Depression is a common but serious mood disorder that affects how a person feels, thinks, and handles daily activities.⁴ More than 90% of people who died by suicide had mental health problems, most notably depression, but most depressed patients do not die by suicide, and many of them never experience SI even in the most severe depressing scenario (Pompili 2019).

Suicide is preventable with timely detection and intervention (Wasserman et al. 2021). As a psychiatric emergency, SI requires immediate attention and assistance from mental health professionals. Delay in detection and assistance may lead to adverse and irreversible consequences. Thus, it is critical to identify those who are at an immediate risk and provide timely intervention (Adrian and Lyon 2018). Unfortunately, like Raimundo, many people with SI do not actively seek help from mental health professionals due to social stigma and other concerns. Standard clinical practice for SI detection (SID) relies on psychological measurements (e.g., Suicidal Ideation Questionnaire Jr. (SIQ-JR) and Suicidal Behaviors Questionnaire Revised (SBQ-R)) and face-to-face interviews, which are passive and not always effective. McHugh et al. (2019) found that 80% of patients who did not receive psychiatric treatment and died from suicide did not report their suicidal thoughts when asked by a practitioner.

Research has shown that some individuals with SI make use of social media to express their suicidal thoughts and intentions, making social media a viable and cost-effective venue for the detection and monitoring of people with SI (Aladağ et al. 2018, Ji et al. 2022b). The extensive literature in psychology and communication suggests that the characteristics of text composed by individuals can reflect their mental states (Park et al. 2013, Calvo et al. 2017). The written text of people who commit or attempt to commit suicide has demonstrated distinctive characteristics that

reflect their suicidal tendencies (Guan et al. 2015, Ji et al. 2022b). Several major social media platforms (e.g., Facebook and Twitter (i.e., X)) have deployed user-initiated mechanisms to monitor SI.⁵ The massive volume of user-generated content, however, poses significant challenges to timely detection. Thus, it is deemed promising and beneficial to develop text analytics-based solutions for automatic SID from social media content.

Current SID research has a few limitations. First, many models use traditional machine learning methods (e.g., decision tree and logistic regression), which require labor-intensive and ad hoc feature engineering (Choudhury et al. 2016, Bitew et al. 2019). Although recent studies have explored deep learning techniques and yielded promising results (Tadesse et al. 2020, Ji et al. 2022b), those studies have primarily adopted conventional deep learning models with static text representations. Transformer-based models such as Bidirectional Encoder Representations from Transformers (BERT) have demonstrated superior performance in a wide range of natural language processing (NLP) tasks, yet remain understudied for SID. Second, decades of studies on NLP tasks have demonstrated the important role of domain knowledge (e.g., domain lexicons) in improving task performance. However, the majority of existing models for SID are solely data driven and overlook domain knowledge. Several recent studies on SID have incorporated lexicons into deep learning models (Allen et al. 2019, Matero et al. 2019, Ji et al. 2022b), but used generic or peripheral lexicons (e.g., general sentiment lexicons) rather than core domain-specific lexicons (see details in Section 2.4). Despite a few studies using suicide-related keyword lists (Huang et al. 2007, Varathan and Talib 2014) or lexicons (Gaur et al. 2019), none of these keyword lists or lexicons was designed specifically for social media content. Notably, those studies primarily matched the terms in social media posts to the keywords or lexicon terms to determine whether a post contained SI, which can be ineffective. Third, most existing models concatenate lexicons with text embeddings learned from pretrained language models. Nevertheless, the embeddings obtained directly from those pretrained models may not align with those of lexicon terms in terms of their representation granularity. Fourth, to the best of our knowledge, few existing studies on SID have evaluated model performance across multiple data sets in different languages at both user and post levels, raising uncertainty about model robustness and generalizability. These research gaps and challenges motivate this study to design novel and effective information technology (IT) artifacts for incorporating domain knowledge into a transformer model for improving SID.

This study proposes a novel knowledge-enhanced transformer-based approach (KETCH) to SID from social media content. Drawing on transformer-based

deep learning, KETCH addresses the limitations of existing methods by extracting dynamic embeddings from transformer models, enhancing them with domain knowledge, integrating text embeddings and knowledge embeddings into a unified framework through model alignment, and adjusting attention weights based on the proposed notion of term-lexicon similarity and term contextual importance. We rigorously evaluate the effectiveness of KETCH by comparing its performance in SID with that of a dozen state-of-the-art baseline models, checking its robustness by developing and testing it at both post and user levels using social media posts in two different languages, examining the impacts of individual design artifacts and different types of lexicons on model performance, assessing the generalizability of KETCH by applying it to depression detection, and exploring the feasibility and user perceptions of KETCH-enabled social media-based intervention via a field study.

2. Background and Related Work

Researchers have identified a wide variety of common causes of SI, such as depression, bullying, discrimination, drug and substance abuse, bereavement, ending a relationship, chronic physical pain or illness, disability, psychache, financial problems, homophobia, feelings of inadequacy or failure, feelings of being burdensome, addiction, concern for the future, and little hope for life (Lee et al. 2010, Bakken and Gunter 2012, Patterson and Holden 2012, Nielsen et al. 2015, Klonsky et al. 2016, Kim et al. 2021). This section reviews four streams of related work, including traditional clinical methods for suicide risk assessment, SID from text using traditional machine learning and deep learning techniques, user-level versus post-level models, and the incorporation of knowledge into SID models.

2.1. Traditional Clinical Methods for SID

Clinical or psychiatric professionals commonly employ interviews or surveys for SID (e.g., Patient Health Questionnaire (PHQ-9); Kroenke et al. 2001). Although this type of method is individual oriented, it often results in a delay between the occurrence of suicidal ideation and its detection (McCarthy 2010), leading to missed opportunities for timely intervention. Additionally, an interview or survey can only take place when an individual actively reaches out for help. Unfortunately, most individuals with SI do not seek help from mental health professionals for various reasons, such as social stigma, fear of trouble, or even unawareness of the need for help (Coppersmith et al. 2018). On average, only 25% of people attempting suicide have visited mental health professionals the week before their attempts (Coppersmith et al. 2018). As a result, individuals with SI cannot be relied upon to take the initiative for mental health

assessment and intervention. Another concern is that interviews or surveys assess SI based on subjective responses from individuals who may not disclose their suicidal thoughts (Gnambs and Kaspar 2015).

2.2. SID from Social Media Text

Numerous studies have shown that people with mental disorders spend a substantial amount of time on social media websites and are willing to disclose their emotions and seek social support due to the reduced fear of social stigma attributable to the anonymity of social media platforms (Coppersmith et al. 2015, Santesteban-Echarri et al. 2017, Yates et al. 2017). User-generated social media content corresponds to a user's behaviors and intentions (Adrian and Lyon 2018). In particular, an individual's language can reflect his/her mood and mental status (Choudhury et al. 2016, Ji et al. 2019a). For example, suicide victims' writing can be more self-focused and may express significantly more unbearable psychological pain and cognitive constriction than the writing of nonsuicidal individuals (Barak and Miron 2005). Therefore, user-generated social media content is a promising source for SID.

Ji et al. (2021) categorized text-based SID methods into content analysis, feature engineering, and deep learning categories. Content analysis methods include lexicon-based filtering (Huang et al. 2007, Varathan and Talib 2014), statistical linguistic features, and topic modeling within suicide-related posts; feature engineering methods refer to the use of traditional machine learning methods and features of user behavior and general text; and deep learning methods employ automatic feature learning. Prior SID studies have exploited text features of social media posts. We group those features into several categories, including word statistics, morphological features, sentiment, topics, and embeddings. Sample word statistics-based features are n-grams (Vioules et al. 2018, Rissola et al. 2019) and term frequency-inverse document frequency (TF-IDF) (Aladağ et al. 2018, Sawhney et al. 2018). Typical morphological features include parts of speech (Choudhury et al. 2016). Sentiment features have also been widely used because individuals with SI tend to express negative emotions (Aladağ et al. 2018, Matero et al. 2019). Topics or latent topics are represented as clusters of similar words or terms, which are commonly extracted from text by topic modeling techniques such as LDA (Blei et al. 2003). Word embedding encodes a word as a real-valued vector in a high-dimensional space. Such a representation allows words with similar semantic meanings to be close to each other in a high-dimensional vector space. Compared with other types of features, which are often manually selected and ad hoc (Zhou et al. 2017), embeddings can be learned from a large corpus automatically and are an integral part of deep learning for text analysis.

Deep neural networks can overcome the inherent limitations of traditional machine learning techniques that rely on feature engineering by learning text representations automatically (Mikolov et al. 2013), and have demonstrated superior performance in a variety of NLP applications (Annapragada et al. 2021, Li and Xing 2021). There is an emerging effort to develop deep learning models for SID. For example, Ji et al. (2019b) trained CNNs and long short-term memory networks (LSTM) to learn latent features of user-generated content; Morales et al. (2019) trained CNN, LSTM, and neural network synthesis models for suicide risk assessment; and Cao et al. (2020) unified a high-level personal suicide-oriented knowledge graph based on the follower-followee relationship discovered from social media sites with a deep neural network for SID. Schoene et al. (2021) extended a standard LSTM for detecting suicide notes by providing fixed boundaries to each layer to reduce the number of hidden units per sub-LSTM hierarchically. Li et al. (2022) proposed a deep hierarchical ensemble model consisting of CNNs for detecting suicide-prone users. Haque et al. (2022) developed deep learning models, such as Bi-LSTM, LSTM, and Bi-GRU, as well as traditional machine learning models. Their evaluation showed that deep learning models outperformed traditional machine learning models.

One major limitation of the previously mentioned deep neural network models is their use of static word embeddings. Word embeddings can be classified into static embeddings such as Word2Vec (Mikolov et al. 2013), where a fixed embedding vector is learned for each word, and dynamic/contextual embedding such as BERT, where an embedding vector for each word varies with its context. Static word embedding models suffer from three major limitations. First, static word embedding relies on word co-occurrences, resulting in fixed vector representations of the same words regardless of the context in which words appear. This fails to address polysemous words whose meanings depend on the context. Second, static word embedding represents each word as an independent vector, providing no shared representations at a subword level. As a result, the representations of two words that are morphologically similar may be considered similar to each other despite being semantically different. Third, static word embedding has difficulty in handling out-of-vocabulary words.

A few recent studies on SID have leveraged pretrained transformer models. For example, Ji et al. (2022a) trained SID models using BERT (Devlin et al. 2019) and RoBERTa (Robustly Optimized BERT Pretraining Approach) (Liu et al. 2019) and tested them using the CLPsych data set (Zirikly et al. 2019). Both models achieved recall and F1 scores in the range of 62.63%–70.33%. Matero et al. (2019) combined BERT-based features and theoretical linguistic

dimensions of text when building detection models. Haque et al. (2020) used two different word-embedding techniques, the GloVe word embedding and transformers including BERT, ALBERT, RoBERTa, and XLNet. The GloVe word embedding was used with a Bi-LSTM model, and the transformers were used with a fully connected layer for classification. These studies, however, extracted embeddings from the pretrained models directly without enhancing them. Moreover, none of them incorporated domain-specific knowledge. For example, even though Matero et al. (2019) combined BERT embeddings and NRC lexicons, the embeddings and lexicons were statically concatenated at the input level. The question about how to incorporate lexicons to refine pretrained post representations dynamically remains unexplored.

2.3. User-Level vs. Postlevel SID

Based on the unit of analysis, existing models for SID can be categorized into user-level and post-level categories. Postlevel models (Huang et al. 2014, Aladağ et al. 2018, Tadesse et al. 2020, Haque et al. 2022) focus on SID from individual posts, aiming to assess a social media user's mental state at the moment of posting and make near real-time interventions possible. These models commonly use the content of individual posts and user features as predictive features. User-level models (Allen et al. 2019, Bitew et al. 2019) aim to predict the degree of latent suicide risk of a specific user by aggregating his/her past posts as the model input. We have identified three main types of methods for aggregating multiple posts of the same user for SI risk prediction at the user level:

- *Concatenation*: It concatenates the representations (e.g., embedding vectors) of individual posts made by the same users in the order of their posting time at the model input level (Allen et al. 2019, Matero et al. 2019, Li et al. 2022). Postconcatenation can be represented by Equations (1) and (2) (Gaur et al. 2019):

$$post_{u,pi} = V_{u,pi,w1} \oplus V_{u,pi,w2} \oplus V_{u,pi,w3} \cdots \oplus V_{u,pi,wN} \quad (1)$$

$$posts_u = post_{u,p1} \oplus post_{u,p2} \oplus post_{u,p3} \cdots \oplus post_{u,pM} \quad (2)$$

where $post_{u,pi}$ denotes the representation of post p_i of user u ; w_j represents the j th word in the post; $\oplus$ represents the concatenation operation; $V_{u,pi,wj}$ represents the representation of the word j in post i of user u ; $posts_u$ represents the representation of all posts made by u ; and N and M represent the number of words in the post i and the number of posts of u , respectively.

- *Averaging*: This method averages the embedding vectors of individual posts from the same user as the

user representation (Matero et al. 2019). The average embeddings have been widely used in analyzing social media content (Son et al. 2018).

- *Ensemble*: It first deploys multiple post-level models to predict if individual posts of the same user contain SI. Then, it aggregates or ensembles the prediction results of individual models to generate a user’s suicide risk prediction (Coppersmith et al. 2018, Bitew et al. 2019).

Given their intrinsic differences, user-level and post-level models serve different purposes and have different applications. As a result, existing research predominantly concentrates on either post-level or user-level modeling, but rarely on both.

2.4. Integrating Knowledge into Deep Learning Models

2.4.1. Framework for Categorizing Methods for Integrating Knowledge into Deep Learning Models. Some researchers (Gaur et al. 2019, Sarsam et al. 2021, Ji et al. 2022b) have explored incorporating knowledge into text-based deep learning models to improve their performance. Manually compiled suicide-related keyword lists or lexicons can facilitate keyword and phrase filtering (Huang et al. 2007, Ji et al. 2022b). We categorize the existing methods for knowledge integration into three levels, including data (the lowest), embedding, and model levels (the highest), as shown in Figure 1.

- *Level 1: Data-level methods* use knowledge to assist with preprocessing or augmenting the input text and/or its meta-features. In the case of *preprocessing*, knowledge can be used to guide the selection of text data for training or fine-tuning the embeddings so that data can satisfy additional constraints posed by knowledge (Ljubic et al. 2020). *Augmentation* aims to augment input text with knowledge. For instance, Roy and Pan (2021) augmented an input sentence for the pretrained ClinicalBERT by appending a second sentence generated based

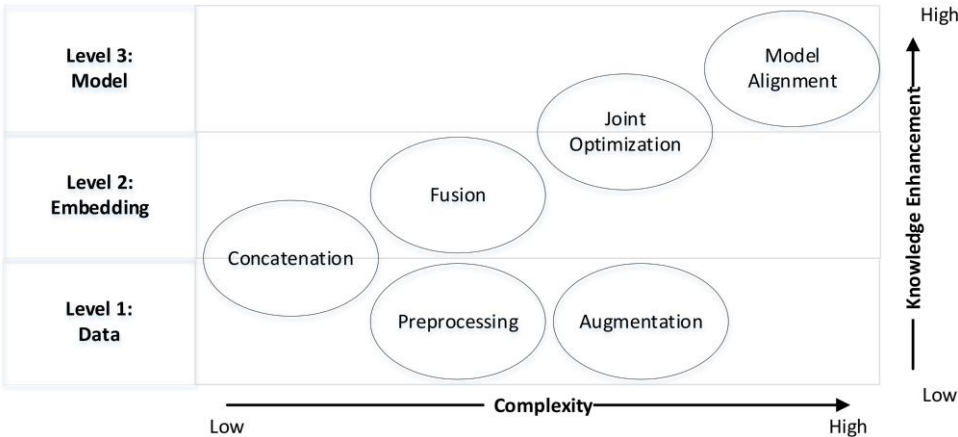
on medical domain knowledge. *Concatenation* at the data level concatenates input text and information from a knowledge source.

- *Level 2: Embedding-level methods* apply knowledge to refine or enrich embeddings. *Concatenation* of multiple embeddings can occur at different levels of granularity (e.g., word- or post-level) and with different model architectures. For example, Roy and Pan (2021) appended embeddings of definitions of the concepts and semantic types based on a knowledge source to the embeddings of an input sentence for relation extraction. *Fusion* integrates pretrained word embeddings with knowledge embeddings. For instance, Choi et al. (2017) used the ICD code hierarchy to improve the initialization of medical concept embeddings, and then fine-tune pretrained embeddings with domain-specific text. *Joint optimization* trains embeddings along with knowledge objectives simultaneously. Changes in a member model are reflected in the joint model, which integrates external knowledge with language models (Roy and Pan 2021).

- *Level 3: Model-level methods* use knowledge as a guide to improve the model architecture design for learning text representations. For instance, a standard BERT model generates text representations at a sentence level, which may restrict its ability to integrate term-level knowledge. To the best of our knowledge, none of the existing models has implemented this type of knowledge integration method.

Within each of the three levels, we introduce complexity as a secondary dimension to further differentiate knowledge-based enhancement methods. For instance, among data-level methods, augmentation is more complex than preprocessing because it involves generating new inputs rather than extracting information from existing data. Generating new data can easily introduce noise into the inputs if the process is not designed well. For embedding-level methods, joint optimization

Figure 1. A Categorization Framework for Knowledge Integration Methods in Deep Learning



is more complex than fusion because the former involves jointly optimizing both knowledge and embedding. However, both data- and embedding-level methods use the output of standard embedding models. This study makes the first attempt at designing a model-level method for integrating knowledge and embeddings.

2.4.2. Incorporating Lexicons into SID Models. One limitation of transformer-based models is their lack of domain knowledge associated with a specific task because transformers are trained with large open corpora. This may lead to ineffective performance of downstream applications (Podkorytov et al. 2021). Lexicons have been used as knowledge in support of many NLP tasks, such as ontology learning (Asim et al. 2018) and sentiment analysis (Dhaoui et al. 2017). Numerous studies have demonstrated the positive roles of lexical knowledge in text classification tasks (Bandhakavi et al. 2017, Le et al. 2019). Therefore, effectively incorporating domain knowledge in the form of an SI lexicon into a transformer-based SID model should enhance its performance. Thus far, only a small number of studies have leveraged lexicons to facilitate SID.

To gain a more complete understanding of extant methods for incorporating lexicons into models, we create a schema to characterize the methods by including studies that used lexicons in various downstream NLP tasks, rather than solely focusing on the SID literature. The schema consists of five key elements: lexicon type, lexicon name, embedding type, integration approach, and modeling techniques. *Lexicon type* is characterized by two dimensions: (1) general versus domain specific and (2) core versus peripheral. A general lexicon typically consists of the vocabulary of a given language and lacks the semantics specific to individual domains or tasks. In contrast, a domain lexicon consists of a vocabulary in a specific domain (e.g., SID). A core lexicon covers the key vocabulary of a knowledge body, whereas a peripheral lexicon includes the vocabulary used attributively to a core vocabulary. The distinction between core and peripheral lexicons lies in their importance and relevance to specific tasks. For example, a sentiment lexicon is peripheral to the task of SID, but is a core lexicon for sentiment analysis. *Embedding type* is classified into static, dynamic, and no embedding. *Integration approach* draws on the categorization framework for integrating knowledge into deep learning models as presented in Table 1 (N/A indicates no embedding or representation learning). Finally, *modeling techniques* refer to the specific machine learning or rule-based techniques used to build detection models.

Some studies have already leveraged lexicons in support of SID, showing the increasing interest of researchers in leveraging lexical knowledge. Drawing on the

schema, we categorize existing SID models that incorporate lexicons in Table 1 and identify several major research gaps. First, most prior studies used peripheral lexicons (e.g., sentiment lexicons and Linguistic Inquiry and Word Count (LIWC)). For example, Ji et al. (2022b) used a general sentiment lexicon to detect individuals' emotions for SID. Only a few studies (Huang et al. 2007, 2014; Gaur et al. 2019) used a suicide-related dictionary or lexicon. Huang et al. (2007) created a dictionary of suicide-related keywords using a variety of online resources, including information from suicide prevention websites and actual suicide notes, which may not reflect or even cover the expressions of social media users in describing their suicidal thoughts. Huang et al. (2014) used a large psychological lexicon and emotion words; nevertheless, the lexicon was not specifically focused on SI or suicide, which could inflate false positives. Gaur et al. (2019) developed a suicide lexicon consisting of medical terminology (e.g., *senile dementia with depressive or paranoid features*) drawn from existing medical resources, which may suffer from the out-of-vocabulary problem when dealing with social media content. The aforementioned studies that involved suicide-related lexicons or dictionaries all used keyword-matching or rule-based models except for Huang et al. (2014), which used traditional machine learning models. For example, Gaur et al. (2019) built a rule-based model using exact and soft matches between the terms in a post and those in their lexicon.

Second, all our reviewed studies except for Matero et al. (2019) used static word embeddings. Matero et al. (2019) developed a logistic regression model using three types of features as the input, including theoretical dimensions (e.g., affect, intensity, and valence of the posts), open-vocabulary features, and meta-features, and a dual RNN architecture using BERT embeddings as the input. Nevertheless, because BERT and other transformer-based models are trained with large open corpora, those models lack domain knowledge associated with a specific task, which may lead to undesirable performance in downstream applications (Podkorytov et al. 2021). Thus, we expect that incorporating a domain lexicon may help.

Third, previous studies mainly concatenated lexicon terms at data and embedding levels as the knowledge integration method. For example, Rissola et al. (2019) concatenated LIWC features with n-grams, word embeddings, and statistical features of users' posting behaviors. None of these studies has explored enhancing text representation models with the guide of a lexicon to improve detection performance, nor has attempted to integrate domain knowledge into term-level embeddings. Moreover, existing work has yet to consider both the contextual importance and lexicon relevance of terms in social media posts and integrate

Table 1. Summary of Existing Work That Incorporates Lexicons for SID

Lexicon types	Lexicons	Embedding type	Integration approach	Modeling techniques	Studies
General/ peripheral lexicons	NRC Affect Intensity Lexicon; SentiStrength	No embedding	Preprocessing (data level)	A semisupervised learning method based on Random Forest (RF)	Sarsam et al. (2021)
	LIWC (Linguistic Inquiry and Word Count)	Static embedding (GloVe)	Concatenation (data level)	Convolutional neural network (CNN)	Allen et al. (2019)
	LIWC	Static embedding (GloVe)	Concatenation (data level)	Logistic regression, support vector machines, and decision trees	Rissola et al. (2019)
	Sentiment lexicons	Static	Concatenation (embedding level)	An attentive relation network	Ji et al. (2022a)
	LIWC, NRC Word- Emotion Association Lexicon, Mental Disease Lexicon	Static	Concatenation (data level)	SVM	Shing et al. (2018)
Suicide-specific/ core lexicons	NRC Affect Intensity Lexicon	Dynamic	Concatenation (data level)	LSTM, GRU, LR, etc.	Matero et al. (2019)
	One dictionary with exact phrases and another wildcard dictionary of suicide-related keywords	No embedding	N/A	Rule-based model based on keyword count	Huang et al. (2007)
	Psychological dictionary and HowNet	No embedding	N/A	Logistic regression, Naïve Bayes, SVM, RF	Huang et al. (2014)
	Suicide Severity Lexicon Social media-based suicidal ideation lexicon	Static Dynamic	N/A Model alignment	Rule-based model Transformer	Gaur et al. (2019) This study

them into the same framework to determine term importance for SID.

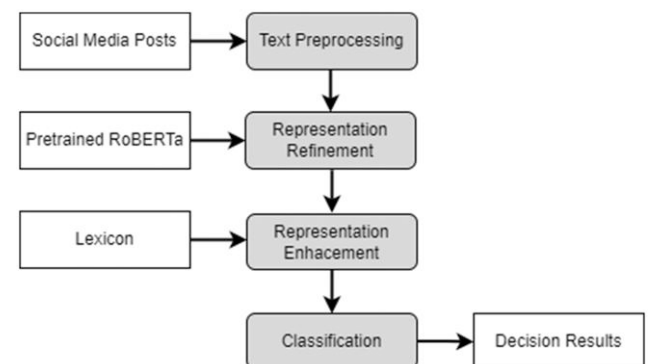
Last, few existing studies have evaluated their proposed models using social media posts in different languages, resulting in a lack of sufficient empirical evidence for the robustness and generalizability of SID models. SID from social media content is a relatively new field, and research on integrating lexicons into deep learning models is still in its infancy.

3. Knowledge-Enhanced Transformer-Based Approach

To fill the previously mentioned research gaps, we propose a novel KETCH for SID from social media text. The design of KETCH is informed by several key design requirements of representation learning, including (1) representing text with dynamic embeddings from a state-of-the-art transformer-based model; (2) integrating structured knowledge (i.e., lexicon) and knowledge extracted from unstructured data (i.e., text) to enhance representation learning; (3) designing a model-level knowledge integration method to enhance representation learning with a domain lexicon; and (4)

considering both domain relevance and contextual importance in determining term importance for SID.

KETCH comprises four main components: text preprocessing, representation refinement, representation enhancement, and classification (Figure 2). Among them, representation refinement and enhancement exemplify embedding- and model-level knowledge integration, and more specifically, knowledge fusion and model alignment, respectively. KETCH first leverages RoBERTa (Liu

Figure 2. The Overall Framework of KETCH

et al. 2019) to learn general text representation, then refines and enhances it to improve SID. KETCH contains two main novel design artifacts: the proposed model alignment for enhancing representation learning with a domain lexicon and the term importance schema considering both domain relevance and contextual importance. KETCH deploys the Tree-based Pipeline Optimization Tool (TPOT), an automated machine learning technique, to select the optimal model based on classification performance automatically.

3.1. Text Preprocessing

Text preprocessing involves data cleansing and tokenization. The latter is language dependent. For instance, Chinese tokenization involves segmenting a consecutive sequence of Chinese characters into words because Chinese written text does not have spaces between adjacent words, while English tokenization typically involves detaching punctuation from words.

3.2. Representation Refinement via Knowledge Fusion

Automatically learned text representations tend to be more robust than manually selected feature representations (Sawhney et al. 2018). RoBERTa is built on the language masking strategy of BERT. BERT applies bidirectional training of transformers and relies on a self-attention mechanism to learn contextual relations among words in the text. A RoBERTa model produces vectors of hidden states as text representations using the self-attention mechanism, which considers the context of social media posts beyond the simple average of word encodings used in static embedding models (e.g., word2vec). RoBERTa reads an entire sequence of words at once to predict a masked word based on the context around it. As it is pretrained on extensive text corpora, RoBERTa effectively addresses the out-of-vocabulary problem.

KETCH leverages RoBERTa's self-attention mechanism to learn the representation of social media posts. Given a collection of posts C , each post $p \in C$ contains M tokens (i.e., $p = \{t_1, t_2, \dots, t_M\}$) and M is the maximum length of a post. We denote M as the median length of all the posts in C , and apply truncation for longer posts and padding for shorter ones. We use special tokens [CLS] and [EOS] to denote the beginning and end of a post, respectively, and [SEP] to denote the delimiter between sentences. The content of p is represented as a vector of the final hidden state $v_p = e_{CLS} \in R^h$, where h is the embedding size and R represents an h -dimensional real-valued space. Following Liu et al. (2019), we set h to 1,024.

Given that text representation from the pretrained RoBERTa model is generic and may not capture linguistic intricacies in the SI text, we refine the text representation with domain-specific text. To this end, we

update the parameters of the RoBERTa model using social media posts with and without SI. Following the RoBERTa encoder layers, we use a fully connected layer with the softmax activation function to classify post p as either having SI or not (see Equation (3)):

$$P(y = 1 | v_p) = \text{softmax}(Wv_p) = W \times \frac{e^{v_p[1]}}{\sum_y e^{v_p[y]}} \quad (3)$$

where v_p is the representation of p and W is a matrix of task-specific parameters. W is determined by using negative log-likelihood as the optimization function (see Equation (4)). Accordingly, the goal of training is to minimize the loss function $J(X)$ by updating W .

$$J(X) = -\log(P(v_p)) \quad (4)$$

Finally, the optimal values of W are used to update the RoBERTa model to improve text representation in its relevance to SI. We use the batch size of eight, weight decay of 0.1, the maximum of four training epochs, and linear learning rate decay with a warm-up ratio of 0.06, which are all empirically determined.

3.3. Representation Enhancement via Model Alignment

3.3.1. Social Media-Oriented Lexicon Construction. The construction of a social media-oriented and expert-curated domain lexicon comprises five main steps:

(1) *Defining the target domain and collecting a representative corpus:* the initial step of lexicon construction involves defining the scope and domain of the lexicon and collecting a corpus of representative social media posts from the domain. The corpus can be collected by web crawlers or similar tools.

(2) *Identifying seed terms:* this is a critical step that can leverage human expertise. This process inherently relies on experts' deep understanding of the domain's subject matter, enabling the identification of seed terms that are not only relevant but also central to the domain's discourse. In addition, specialized training for experts in areas, such as subject matter knowledge, can be important for ensuring the quality of the seed terms. The experts can first label the data samples in the corpus based on whether individual samples are related to the target problem or not, then identify as many quality seed terms from the subset related to the target problem as possible.

(3) *Refining seed terms:* the identified seed terms are evaluated and those that are peripheral or tangential to the domain's core subject matter may be removed. This step can again heavily rely on the domain knowledge of human experts.

(4) *Expanding seed terms:* this step will expand the list of refined seed terms by adding a broader array of terms that capture the linguistic diversity and semantic

richness of the domain, ensuring a comprehensive coverage of its lexical landscape. It involves identifying and incorporating additional terms that exhibit high semantic similarity to the refined seed terms. Unlike the previous steps, the enrichment of seed terms can largely be automated, using computational methods to discover terms with similar meanings or contexts within the domain.

(5) *Pruning the terms*: human experts conduct a thorough review of the expanded list of terms and eliminate those that are repetitive, less important, or less relevant to the target domain's subject matter to improve the parsimony of the lexicon while maintaining sufficiency.

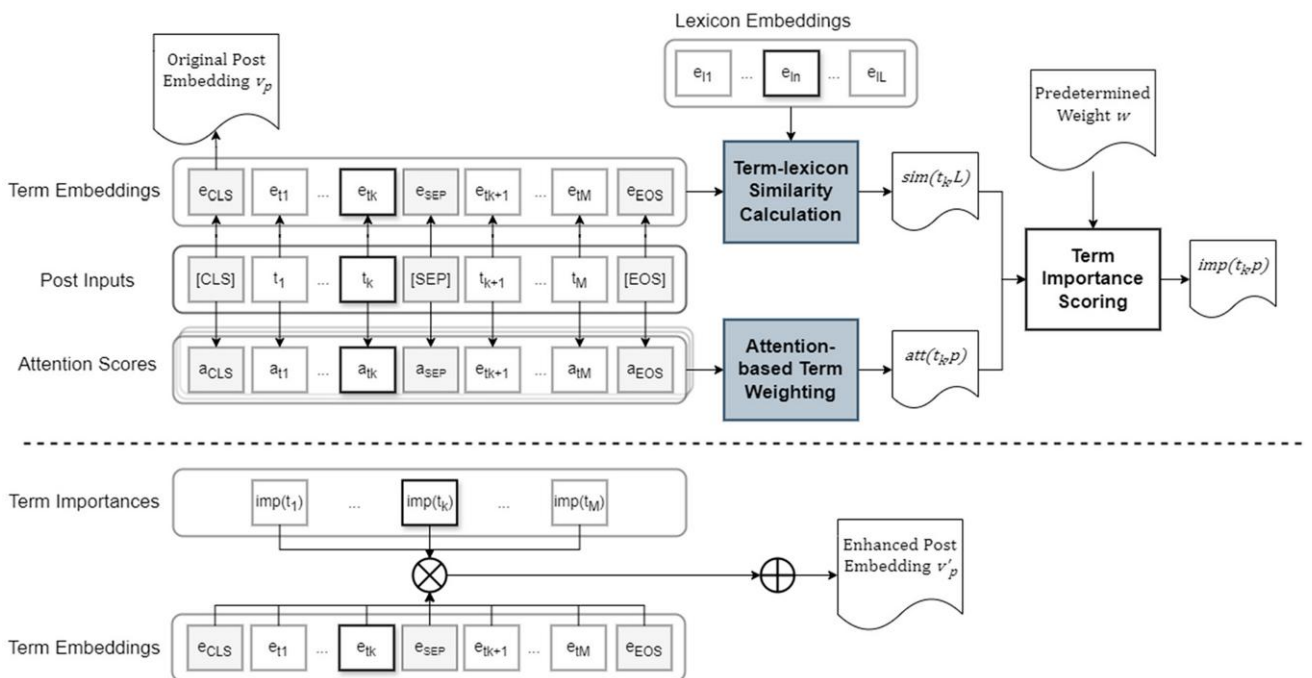
Domain experts play an important role in the process of lexicon construction to ensure the accuracy and quality of the selected lexicon terms. Nevertheless, human decision making involves individual biases (Johnson 2021). Thus, it is advisable for those lexicon construction steps relying on human expert judgment to involve multiple experts and adopt reliability measures such as interrater reliability (e.g., 80% or higher agreement) or majority vote to minimize potential individual biases.

3.3.2. Enhanced Representation Learning. We incorporate the developed domain lexicon into the fine-tuned RoBERTa model to enhance representation learning. In particular, our design of enhanced representation learning is motivated by two key considerations. One is that a term in a social media post that is more

similar to an SI lexicon term should generally signal higher relevancy to SI. The other consideration is that the meanings of a lexicon term or potential signifier terms of SI may vary in different contexts. For example, although “杀” (kill) is an SI lexicon term, it does not necessarily indicate suicidal ideation in a certain context, such as “在这所医院,杀菌是通过一个化学过程来完成的。” (“In this hospital, sterilization is accomplished by a chemical process.”) (See Appendix B for additional sample social media posts.) Therefore, we posit that only considering whether a social media term exists in the SI lexicon is insufficient for making a correct detection decision. Lexicon (signifier) terms that are more important to the context of a social media post should convey stronger signals of post relevancy to SI. To this end, we design a new artifact, $imp(t, p)$, to determine the importance of term (or token) t in social media post p to the prediction of SI.

We operationalize the above considerations by introducing a novel model alignment method for knowledge enhancement. In view that the unit of a lexicon is a term, whereas the standard representation learning output of a transformer-based model is at the sentence/post level, we propose to align the two by using the lexicon as a guide in representation learning. Our proposed model alignment method consists of two key components: aligned dynamic embedding and lexicon-based enhancement, which collectively measure $imp(t, p)$ (Figure 3). The top half of the figure illustrates how we derive the aligned dynamic embedding

Figure 3. (Color online) Enhanced Representation Learning



via attention-based term weighting $att(t, p)$ and how we operationalize lexicon-based enhancement via term-lexicon similarity $sim(t, L)$. The bottom half of the figure shows how we generate an enhanced representation of p as v'_p by integrating the term-lexicon similarity and attention-based term weighting in $imp(t, p)$.

Aligned dynamic embedding for term representation determines the contextual importance of term t to post p . Not all terms t_i ($i \in 1 \dots M$) in p are equally important. Aligned dynamic embedding draws on the self-attention mechanism from a transformer-based model but is different from the latter. The most important difference is that a transformer model, including its fine-tuned counterpart, essentially derives attention scores at a post level in the output representation by default. Formally, a transformer model represents p as a real-valued vector of hidden states, $v_p \in R^h$, captured by the output embedding of the [CLS] token in the embedding space R^h , where h is the embedding size. Such a representation is limited when it comes to measuring term-level information such as attention and similarity because it is difficult to use the post-level attention to differentiate terms in the same post, and more importantly, the representation does not capture the dynamic attention of t to other terms in the same post p .

To address the above limitation, the aligned dynamic embedding drills down to the level of individual terms by extending term (or token) embeddings to incorporate dynamic term-level attention in p , $att(t, p)$. As such, the aligned dynamic embedding enables the delineation of different terms based on their importance to post p . To this end, we extract the embedding of term t , $v_t = (v_t^1, \dots, v_t^h) \in R^h$, from RoBERTa and represent it as an h -dimensional real-valued vector, where v_t^i ($i \in 1 \dots h$) is a scalar value. To enable dynamic term embedding, we take other terms in p into consideration when determining the weight of t . Moreover, moving beyond the norm of only using the outputs of the last layer in a transformer model as representations, we extract attention scores from the last q layers ($1 < q < Q$, where Q is the total number of layers in RoBERTa), which enables better contextualization of the term representation via modifying its embeddings to reflect different meanings of the same term in different contexts (i.e., contextual importance). Accordingly, $att(t, p)$ is defined as the average attention score across all n attention heads within the last q layers of the RoBERTa model given by Equation (5):

$$att(t, p) = \sum_{i=1}^q \sum_{j=1}^n \frac{a_{ij}(t, p)}{q \times n} \quad (5)$$

where $a_{ij}(t, p)$ is the raw self-attention score of t in p on attention head j in layer i ; $q \leq Q$, and both Q and $n = 12$

in RoBERTa. The optimal value of q , which is four, is determined empirically.

Lexicon-based enhancement incorporates the lexicon to determine the relevance of term t to SI. It is measured by term-lexicon similarity, $sim(t, L)$, where $t \in p$ and L denotes the SI lexicon. The value of $sim(t, L)$ is derived based on the similarity measurement between t and lexicon term $l \in L$, $sim(t, l)$, where l is represented as an h -dimensional real-valued vector $v_l = (v_l^1, \dots, v_l^h)$ in the same embedding space as t . We choose cosine similarity because it is one of the most commonly used similarity metrics. The term-lexicon similarity of t is defined as the maximum similarity between t and l , as shown in Equation (6).

$$sim(t, L) = \max_{l \in L} sim(t, l) \quad (6)$$

The overall importance of t to SI in p , $imp(t, p) \in (0, 1)$, is defined as a weighted sum of term-lexicon similarity and dynamic term-level attention (see Equation (7)):

$$imp(t, p) = w \times sim(t, L) + (1 - w) \times att(t, p) \quad (7)$$

where $w \in [0, 1]$ denotes the weight. After enhancing the embeddings of all the terms in p with $imp(t, p)$, we aggregate them to generate an enhanced representation of p , as shown in Equation (8):

$$v'_p = \frac{1}{M} \sum_{i=1}^M (imp(t_i, p) \times v_{t_i}) \quad (8)$$

where v_{t_i} is the original embedding of term t_i and M is the maximum length of post p . When the value of $imp(t_i, p)$ is close to one (i.e., t_i has a high similarity to some lexical terms and/or a high attention score), the relative influence of t on the representation of p is maximized. As such, KETCH not only captures the importance of lexical terms to SID, but also enhances the representations of other nonlexical terms with contextual importance. In the latter case, terms having low semantic similarities with lexicon terms but relatively high contextual importance still contribute to SID.

3.4. Classification

SID can be modeled at the post or user level (see Section 2.3). We primarily focus on post-level modeling in this study for two reasons. First, detecting SI from individual posts enables timely prevention and intervention (Wasserman et al. 2021). A social media user's most recent post best reflects his/her current mental state, which forms the basis of SID. Second, based on the reported team performances of the CLPsych 2019 shared task, incorporating historical posts of the same users did not improve SI detection performance (Zirikly et al. 2019). Morales et al. (2019) also reported "systems trained at the post level outperforming user-

based systems” (p. 180). That being said, we have also built and evaluated KETCH-based user-level models as a robustness check in this study (see Section 5.3).

We build classification models using the enhanced post representations as the input to determine whether a social media post involves SI. Tuning a large number of hyperparameters to achieve optimal performance is a challenge. We adopt TPOT (Olson et al. 2016) for model optimization. TPOT uses machine learning algorithms from the popular Scikit-learn machine learning library and a genetic programming stochastic global search procedure to discover the best hyperparameters and model ensembles given a data set.

4. Evaluation

4.1. Data Collection and Chinese SI Lexicon Construction

We collected two social media datasets for the evaluation of KETCH for SID. The first one was collected from Sina Weibo, the second-largest social media platform in China. Among its users, 50.1% were male; 77.8% had a college degree or higher; and 80% were younger than 30 years old. We crawled 104,219 posts generated by 29,388 users from a large Sina Weibo group associated with an individual (Weibo ID: 走饭 (Zou Fan)⁶) who committed suicide in 2012 due to depression. The Weibo posts were generated between January 1, 2016, and July 24, 2018. After excluding the posts with fewer than five Chinese words, we obtained 99,030 posts, which were randomly stratified into training (89,126 posts, including 14,342 labeled as SI posts) and test data (9,904 posts, including 1,581 labeled as SI posts). A team of six domain experts (e.g., psychiatrists and doctors in a mental health hospital) first manually and independently labeled the collected posts as either indicative of SI or not, with each post labeled by three experts. The interrater reliability measured by Krippendorff’s alpha was 0.83. The experts discussed the posts that received mixed labels and reached a consensus on each. The average length of posts was 31.26 Chinese characters.

The second dataset was collected from the r/SuicideWatch subreddit on Reddit and used in the CLPsych 2019 Shared Task (Zirikly et al. 2019). The dataset consisted of 919 English posts from 496 users for training and 186 posts from 125 users for testing. The average length of Reddit posts was 259.59 English words. The user labels, namely no, low, moderate, or high risk of suicide, were derived based on the initial annotations of individual posts of the same users. To build a binary classifier, this research considers the users labeled as having no suicide risk as “no SI” and those labeled as low, moderate, or high risk as “having SI,” following Zirikly et al. (2019). As a result, the numbers of users with SI in the training and test data

sets were 369 and 93, respectively. The distributions of user count across the varying number of posts and the frequency distributions of posts containing the SI lexicon terms are reported in Appendix C. It is worth noting that it is very common for SID researchers to collect social media posts from online user groups associated with suicide or other related mental health issues (Bitew et al. 2019, Zirikly et al. 2019, Ji et al. 2022b).

We developed a Chinese SI lexicon by following the general procedure for constructing a social media-oriented and expert-curated domain lexicon (see Section 3.3.1). Specifically, in the first step (defining a domain and collecting a corpus), we collected more than 4,600 Sina Weibo posts from a list of confirmed suicidal users and other active users. The latter were randomly selected to enable fair representation of potential variant suicidal expressions. To identify seed terms (Step 2), we recruited a panel of eight experts specialized in suicide research (who were experienced in providing psychiatric consulting services and/or treating patients with suicide risk). The panel first completed a training session on the framework of Rudd et al. (2006) of warning signs for suicide risk and then analyzed 50 randomly selected Weibo posts to identify expressed suicide risk independently. The interrater reliability of the analysis results measured by Krippendorff’s alpha was 0.82. Next, the panel of experts was divided into two four-person groups, each processing half of the Weibo posts to identify those with suicide risk separately. Only those posts that received an agreement from at least three experts in each group were selected, resulting in about 2,000 posts with suicide risk. The same groups of experts were asked to identify terms signaling suicide risk from the selected posts independently. Only terms that were agreed upon by at least three experts in each group were selected as candidate terms. In the refinement step, another three coders specialized in suicide research rated those candidate terms based on their sensitivity to suicide risk ranging from low to high. Those terms identified as having low sensitivity or higher sensitivity to suicide risk only in a very specific context, and so on, and thus were suggested for removal by more than one coder, were discarded. This step eventually resulted in 1,862 seed suicide risk terms. The same three coders were also asked to classify the seed terms into multiple categories based on a pre-developed categorization framework, consisting of SI, (negative) personality, psychache, and so on. The results were confirmed with an agreement of two or more coders. To further expand the seed terms (Step 4), we deployed a corpus-based method that automatically selected four terms having the highest semantic similarity to each of the SI seed terms from a large corpus of Sina Weibo posts, measured by the

cosine similarity of word2vec vector representations. In the last step, the three coders pruned the expanded terms by removing redundant or irrelevant terms to SI based on an agreement of at least two coders, resulting in a total of 586 SI terms. To improve the parsimony of the lexicon while maintaining sufficiency, another three researchers specialized in SID further filtered the terms independently, removing those terms suggested by at least two researchers and verified via group meetings. Some patterns of filtered terms include (1) redundancies caused by repeating the same term (e.g., 很累/很累 很累 (tired/very tired)); (2) redundancies in terms of content words (e.g., 结束/结束了 (finished/it is over)); (3) part-whole redundancies (e.g., 今世, 今生/今生今世 (this life)); (4) terms comprising high-frequency and common words (e.g., 极乐世界 (Nirvana) containing 世界 (world)); (5) long Chinese idioms (e.g., 福无双至, 祸不单行 (Blessings never come in pairs, misfortunes never come alone)); and (6) non-Chinese characters (e.g., fin and goodbye), and so on. We finally obtained an SI lexicon consisting of 320 terms (see Appendix A). The Weibo data set and experts involved in constructing the lexicon are separate from those involved in model training and testing and data annotation.

4.2. Experiment Design

We evaluated KETCH and its main design artifacts empirically and rigorously via four experiments:

- Experiment 1 assessed the KETCH and baseline models with the Chinese Weibo data set, and examined the effects of incorporating the proposed representation refinement and representation enhancement into the baseline RoBERTa model on SID.
- Experiment 2 assessed the relative influence of lexicon-based enhancement and aligned dynamic embedding in representation enhancement of KETCH on SID using a sensitivity analysis, and compared the effectiveness of our SI lexicon, a machine-generated SI lexicon, and a peripheral lexicon.
- Experiment 3 investigated the robustness of KETCH by building user-level models for the prediction of suicidal risk of social media users using the English Reddit data set and comparing them with state-of-the-art baseline models.
- Experiment 4 assessed the generalizability of KETCH with a different domain by building and evaluating a KETCH-based model for depression detection using another social media data set.
- A field study was conducted to assess user perception of a KETCH-enabled SI monitoring and intervention solution, serving as a feasibility analysis of KETCH for near real-time SID.

4.3. Model Setup and Baselines

4.3.1. Data Sampling Strategies. SID is commonly treated as a binary classification problem (i.e., with or

without SI). One common challenge in building classification models is imbalanced training data that can cause the trained models to be biased toward the majority class. In our Sina Weibo data set, the ratio of posts with versus without SI was approximately 1:6, which was highly imbalanced. In this study, we deployed SMOTE (Synthetic Minority Oversampling TEchnique) (Chawla et al. 2002) to generate balanced training data. SMOTE is an oversampling technique that increases the number of minority-class samples by generating new synthetic instances from existing minority cases. We empirically set q in Equation (5) to 4 and w in Equation (7) to 0.6 in the experiments.

4.3.2. Baseline Models and Hyperparameter Tuning.

Our baselines consisted of 12 state-of-the-art SID models (Table 2). All but two are deep learning models because of their superior performance to traditional machine learning models. CNN, LSTM, BiLSTM, GRU, and Bi-GRU are the most commonly used deep learning techniques in existing SID studies (Ji et al. 2019b, Morales et al. 2019, Li et al. 2021, Haque et al. 2022). They mainly use static word embedding (e.g., word2vec) as text representation. We empirically set the embedding size of word2vec to 150 in this study. We also included RoBERTa (RB) as a baseline model, allowing us to examine the incremental effects of incorporating representation refinement and representation enhancement. Furthermore, the baseline models also included state-of-the-art methods for other related mental health problems, such as depression and self-harm (Martínez-Castaño et al. 2020, Bucur et al. 2021). Bucur et al. (2021) used a BERT model, whereas the XML-RoBERTa model (Martínez-Castaño et al. 2020) was a pretrained multilingual language model originally trained for different cross-lingual transfer tasks. We used a fully connected layer as the classifier, considered different model configurations, and reported the performance of the optimal model. The last four baseline models in Table 2 used one or more lexicons. Among them, only the rule-based model (Gaur et al. 2019) used a core suicide lexicon, which, however, was not social media oriented (see Section 2.4). We experimented with their model by varying the cutoff threshold value of cosine similarity between 0.1 and 1 with an increment of 0.1 and chose the optimal value of 1 (i.e., an exact match between a post term and a lexicon term), which yielded the best model performance. NRC VAD contains more than 20,000 English words and their valence, arousal, and dominance scores. LIWC is another peripheral lexicon used by prior studies on SID (Ríssola et al. 2019).

4.4. Evaluation Metrics

We used four metrics, including precision, recall, F1 score, and accuracy, to assess model performance in

Table 2. The Selected State-of-the-Art Baseline Models

Baselines	Deep learning techniques and settings	Feature representations
CNN	Containing two convolutional units, each consisting of a Conv1D layer with 32 neurons, a max pooling layer, and a dropout layer with a dropout ratio of 0.5	Static word embedding
LSTM	Consisting of two LSTM layers with 128 neurons per layer and three dropout layers with a dropout ratio of 0.2	Static word embedding
Bi-LSTM	Bi-LSTM sharing the same architecture as the above	Static word embedding
GRU	Consisting of two GRU layers with 128 neurons per layer and three dropout layers with a dropout ratio of 0.2	Static word embedding
Bi-GRU	Bi-GRU sharing the same architecture as GRU	Static word embedding
RoBERTa (RB)	A RoBERTa model (one fully connected layer with the softmax activation)	Post embedding
BERT (Bucur et al. 2021, Diniz et al. 2022)	A BERT model (one fully connected layer with the softmax activation)	Post embedding
XLM-RoBERTa (Martínez-Castaño et al. 2020)	An XLM-RoBERTa model with one fully connected layer and the softmax activation	Post embedding
Rule-based (Gaur et al. 2019)	Based on the cosine similarity between the embeddings of terms present in a user's content and the embeddings of lexicon terms (empirically selected 0.6 as the cutoff threshold)	Term embedding Lexicon: a medical professional lexicon
Open Theory model (Matero et al. 2019)	Logistic regression (LR)	Open-vocabulary (e.g., BERT embeddings and LDA topics), theoretical (e.g., affect, intensity, and valence scores), and meta-features
DeepAtt (Matero et al. 2019)	LSTM with hierarchical post-level attention	Open-vocabulary features, affect, intensity, and the NRC VAD lexicon scores of each post
ConvSent (Allen et al. 2019)	Stacked with one CNN layer and two fully connected layers; each of the dense layers has 250 nodes.	45 LIWC lexicon features of each post (we employed the simplified Chinese version of LIWC (Zhao et al. 2016))

SID. Precision is defined as the percentage of posts detected as having SI that truly have it; recall is the percentage of posts with SI that are detected correctly; F1 score is a harmonic mean of precision and recall; and accuracy is the number of correct predictions divided by the total number of predictions.

We introduced our training and test set split for the Chinese Weibo data set and the English Reddit data set in Section 4.1. We first trained each model with the training set using five-fold cross-validation for hyperparameter optimization and model selection to prevent overfitting. Then, we evaluated the trained models using the test data set. Moreover, we performed the fivefold cross-validation in 30 independent runs, using a repeated random subsampling validation process for each run. The reported performance values are the average across the 30 runs. We also conducted multiple comparisons in repeated measures analysis with Bonferroni adjustments to compare the SID performance of different models.

5. Results

5.1. Experiment 1: KETCH and Baseline Model Performance

5.1.1. Baseline Model Performance. The performance of all baseline models is reported in Table 3, with the

Table 3. Performance of the Baseline Models

Models	Precision	Recall	F1	Accuracy
CNN	0.420	0.500	0.457	0.840
LSTM	0.512	0.642	0.514	0.858
GRU	0.509	0.609	0.509	0.850
BiLSTM	0.656	0.636	0.637	0.858
BiGRU	0.669	0.699	0.671	0.869
RB	0.747	0.674	0.705	0.906
BERT	0.652	0.690	0.666	0.797
XML-RoBERTa	0.656	0.674	0.671	0.840
Rule-based	0.210	0.965	0.345	0.415
Open Theory	0.689	0.298	0.416	0.866
DeepAtt	0.750	0.710	0.724	0.858
ConvSent	0.682	0.402	0.506	0.875

Note. Values in bold represent the best performance results for each evaluation metric.

highest value of each measure highlighted in bold. First, among all the baseline models, DeepAtt achieves the highest precision and F1; RB achieves the highest accuracy and the second highest precision and F1. Second, RB performs best overall among all the baselines without using any lexicon. RB consistently outperforms the BERT and XML-RoBERTa models across all measures except recall. Third, although the rule-based model achieves the highest recall, it performs far worse than other baseline models in precision, F1, and accuracy. These results suggest that (1) RB is the better choice than BERT, XML-RoBERTa, and other baseline models without a lexicon, and (2) incorporating a lexicon does not necessarily lead to performance improvement. The type of lexicon(s) selected and the method of integrating lexicons may influence model performance.

5.1.2. Performance of KETCH and the Impact of Key Design Artifacts with Weibo Data. To achieve highly contextualized word embeddings, it is suggested that researchers retrieve embeddings from the last n layers rather than the last layer of a transformer model (Ethayarajh 2019). Thus, we performed a sensitivity analysis by varying the value of n ($n = 1, 2, 3$, or 4) of the pretrained RoBERTa model for fine-tuning. The results show that the anisotropy of the representations is maximized when $n = 4$. Thus, we extracted embeddings from the last four layers of the model.

Table 4 reports the results of an ablation experiment that incrementally incorporated representation refinement (RR), representation enhancement (RE), and SMOTE (SM) into the RB baseline model. We performed repeated-measures analysis of variance (ANOVA) by using knowledge enhancement as the independent variable and each of the performance metrics as the dependent variable. The results reveal a significant main effect of knowledge enhancement on precision ($F(3, 87) = 1,615, p < 0.001$), recall ($F(3, 87) = 12,482, p < 0.001$), F1 ($F(3, 87) = 5,398, p < 0.001$), and accuracy ($F(3, 87) = 212, p < 0.001$).

Table 5 reports the results of multiple comparisons of the model performance with Bonferroni adjustments. It shows that incorporating RR improves the performance of the baseline RB model across all the measures ($p < 0.001$ for all measures except for accuracy ($p < 0.01$)). Introducing SM into the RB + RR model

leads to a performance improvement in recall and F1 ($p < 0.001$) but causes a performance decline in precision and accuracy ($p < 0.001$). Finally, integrating RE into the RB + RR + SM model (i.e., KETCH) leads to further improvements in all four measures ($p < 0.001$).

5.2. Experiment 2: Effects of Lexicon-Based Enhancement and Aligned Dynamic Embedding

We conducted a sensitivity analysis on the value of w in Equation (7) to assess the relative importance of the two components of RE in KETCH: lexicon-based enhancement and aligned dynamic embedding.

We varied the value of w between 0 and 1, with an increment of 0.1. When $w = 0$, KETCH only considered aligned dynamic embedding (contextual importance) but not lexicon-based enhancement (lexicon-term similarity) or vice versa when $w = 1$. The results (Table 6) reveal that KETCH achieves the best performance when $w = 0.6$. In other words, models integrating both lexicon-based enhancement and aligned dynamic embedding outperform those incorporating either of them alone. These results suggest that both components of RE contribute to improving the performance of KETCH in SID.

To investigate the comparative efficacy of our expert-curated Chinese SI lexicon with an SI lexicon automatically constructed by a machine learning model or a peripheral lexicon in aiding SID, we developed two alternative post-level KETCH models using the Chinese Weibo data. In each model, we kept the RE component while replacing the expert-curated Chinese SI lexicon with either a ChatGPT-generated Chinese SI lexicon (GPT-CSI) or the Chinese version of the NRC VAD lexicon (Mohammad 2018). We leveraged GPT-3.5 to generate the Chinese SI lexicon by using a prompt “Can you provide a suicidal ideation lexicon containing 320 Chinese terms derived from social media content solely for academic research purposes?” GPT-CSI consisted of the top 320 Chinese terms identified by ChatGPT as most relevant to SI. The results, as reported in column (a) of Table 7, demonstrate that our expert-curated Chinese SI lexicon is more effective than GPT-CSI and NRC VAD lexicons for SID. GPT-CSI, a core lexicon, is more effective than NRC VAD, a peripheral lexicon. Nevertheless, KETCH with the NRC VAD lexicon still achieves a

Table 4. Model Performance in the Ablation Experiment (Means [Standard Deviations])

Models	Precision	Recall	F1	Accuracy
RB	0.747 [0.020]	0.674 [0.021]	0.705 [0.022]	0.906 [0.025]
RB + RR	0.753 [0.018]	0.689 [0.017]	0.717 [0.018]	0.910 [0.020]
RB + RR + SM	0.728 [0.015]	0.857 [0.014]	0.765 [0.016]	0.893 [0.017]
KETCH (RB + RR + SM + RE)	0.826 [0.007]	0.881 [0.009]	0.868 [0.009]	0.935 [0.010]

Notes. Values in bold represent the best performance results for each evaluation metric. RR, representation refinement; RE, representation enhancement; SM, SMOTE.

Table 5. Multiple Pairwise Comparisons of Model Performance

(a) Precision				
Model (I)	Model (J)	Mean difference (I – J)	Standard error	<i>p</i> value
RB + RR	RB	0.006	0.001	<0.001
RB + RR + SM	RB	–0.019	0.001	<0.001
	RB + RR	–0.025	0.000	<0.001
KETCH (RB + RR + SM + RE)	RB	0.079	0.002	<0.001
	RB + RR	0.074	0.002	<0.001
	RB + RR + SM	0.098	0.002	<0.001
(b) Recall				
Model (I)	Model (J)	Mean difference (I – J)	Standard error	<i>p</i> value
RB + RR	RB	0.016	0.001	<0.001
RB + RR + SM	RB	0.183	0.001	<0.001
	RB + RR	0.167	0.001	<0.001
KETCH (RB + RR + SM + RE)	RB	0.207	0.002	<0.001
	RB + RR	0.191	0.002	<0.001
	RB + RR + SM	0.024	0.001	<0.001
(c) F1				
Model (I)	Model (J)	Mean difference (I – J)	Standard error	<i>p</i> value
RB + RR	RB	0.012	0.001	<0.001
RB + RR + SM	RB	0.060	0.001	<0.001
	RB + RR	0.048	0.000	<0.001
KETCH (RB + RR + SM + RE)	RB	0.164	0.002	<0.001
	RB + RR	0.152	0.002	<0.001
	RB + RR + SM	0.103	0.001	<0.001
(d) Accuracy				
Model (I)	Model (J)	Mean difference (I – J)	Standard error	<i>p</i> value
RB + RR	RB	0.004	0.001	<0.01
RB + RR + SM	RB	–0.013	0.002	<0.001
	RB + RR	–0.017	0.001	<0.001
KETCH (RB + RR + SM + RE)	RB	0.029	0.003	<0.001
	RB + RR	0.025	0.002	<0.001
	RB + RR + SM	0.043	0.001	<0.001

Table 6. Results of Sensitivity Analysis on w in Equation (7) with the Weibo Data Set

w	The Sina Weibo data set			
	Precision	Recall	F1	Accuracy
0	0.701	0.685	0.700	0.817
0.1	0.716	0.695	0.701	0.817
0.2	0.729	0.716	0.729	0.835
0.3	0.732	0.716	0.732	0.844
0.4	0.762	0.732	0.741	0.858
0.5	0.789	0.757	0.778	0.910
0.6	0.826	0.881	0.868	0.935
0.7	0.800	0.833	0.830	0.920
0.8	0.762	0.716	0.741	0.858
0.9	0.732	0.732	0.732	0.888
1	0.729	0.732	0.729	0.858

Note. Values in bold represent the best performance results for each evaluation metric.

higher F1 than all baseline models and near-the-top precision and recall among all baseline models (Table 3). Our comparison of the GPT-CSI and our CSI lexicons reveals that 17.5% of their terms partially overlap. The average cosine similarity between lexicon terms in GPT-CSI and their most similar terms in our CSI is 0.7466. The contextual word embeddings employed in KETCH facilitate comparing term similarities based on semantics rather than mere surface string matching. Thus, different terms from these two lexicons may still have a high semantic similarity.

To gain deeper insights into the role of lexicon terms, we analyzed the performance of KETCH in SID from posts with versus without lexicon terms. The results, as shown in Table 8, show that KETCH performs better on posts containing lexicon terms than on those

Table 7. Comparison of Performances of KETCH with Different Lexicons

Performance metrics	(a) Weibo data set			(b) Reddit data set		
	GPT-CSI	NRC VAD	Our CSI	GPT-ESI	NRC VAD	Our ESI
Precision	0.801	0.728	0.826	0.912	0.794	0.923
Recall	0.863	0.765	0.881	0.943	0.787	0.955
F1	0.832	0.737	0.868	0.927	0.791	0.939
Accuracy	0.897	0.833	0.935	0.896	0.876	0.912

Notes. Values in bold represent the best performance results for each evaluation metric. CSI, Chinese suicidal ideation lexicon; ESI, English suicidal ideation lexicon.

without lexicon terms in all performance measures, yet its performance on the latter is still reasonably good and consistent.

5.3. Experiment 3: KETCH-Based User-Level Models for Suicide Risk Prediction: A Robustness Check

As discussed in Section 2.3, user-level models have a different aim from post-level models. To examine whether our methodological innovations are generalizable to user-level models, we developed and evaluated a number of KETCH-based user-level models for predicting the suicide risk of social media users using the English Reddit data set (see Section 4.1) under the following different settings:

5.3.1. Different Lexicons. To investigate the impact of different lexicons on model performance, we developed two lexicons for this experiment due to the lack of existing social media-oriented English SI lexicons. The first lexicon, referred to as ESI, was created by drawing terms from two sources: the English translation of our Chinese SI lexicon and the suicide risk severity lexicon used in the study of Gaur et al. (2019). Two researchers who were actively engaged in research on suicide risk detection from social media content and proficient in both English and Chinese independently translated our Chinese SI lexicon into English, then discussed different translations and selected the more suitable ones. Given that complex professional medical terms (e.g., “global deterioration scale finding for assessment of primary degenerative dementia”) are unlikely to appear in user-generated social media posts, those complex terms were excluded yet partial key terms (e.g., “dementia”) may be selected in the lexicon. In the end,

we obtained an ESI lexicon of 370 terms. The second English SI lexicon, referred to as GPT-ESI, was generated by ChatGPT using a prompt similar to the one used for creating the GPT-CSI lexicon, with the exceptions of specifying 370 terms and removing “Chinese” from the prompt. The average cosine similarity between lexicon terms in GPT-ESI and their most similar terms in ESI is 0.6671. Additionally, we also used the NRC VAD English lexicon.

5.3.2. Methods for Aggregation of Multiple Posts from the Same User. To understand the potential impact of different methods for aggregating multiple posts at the user level, we developed user-level models using three different post aggregation methods, namely concatenation, averaging, and ensemble methods (see Section 2.3). Specifically, the ensemble method predicted a user to have suicide risk if any of his/her last N posts is detected to have SI, following the approach of Shing et al. (2018).

5.3.3. Number of Posts. To investigate the potential impact of the number of posts on model performance, we developed different models by varying the number of a user’s most recent posts, ranging from one to five posts, as the input and applied paddings for users with fewer posts in the data set. We used up to five posts from the same user for two major reasons. First, 480 (77.3%) users in the Reddit data set made only one post, and the average number of posts per user was 1.66. Second, there are currently no established standards or guidelines regarding the optimal number of aggregated posts. For example, the BERT-multiple-instance-learning model (Ambalavanan et al. 2019) achieved a better flagged F1 measure score when using five posts than using other numbers of posts; and the CLaC system (Mohammadi et al. 2019), the best-performing model in Task A of the CLPsych 2019 shared task, used the first 200 or 400 tokens (equivalent to about five posts based on the average size per post) when feeding the posts to submodels.

Combining the previously mentioned design choices resulted in a total of 45 KETCH-based user-level models. The performances of these models reveal that when

Table 8. Performance of KETCH Models on Posts with vs. without Lexicon Terms

With or without lexicon terms	Precision	Recall	F1	Accuracy
With	0.840	0.882	0.870	0.939
Without	0.817	0.878	0.856	0.926

using our ESI, KETCH with averaging the five most recent posts achieves the best performance. Meanwhile, when using GPT-ESI, KETCH with the ensemble method on the five most recent posts yields the best overall detection performance. Comparing the best performances of KETCH with three different lexicons on the Reddit data set, as presented in column (b) of Table 7, reveals that KETCH with our ESI lexicon outperforms both KETCH with GPT-ESI and KETCH with NRC VAD, and KETCH with GPT-ESI outperforms KETCH with NRC VAD. Furthermore, the performance difference between KETCH with ESI and KETCH with GPT-ESI is much smaller than that between either model and KETCH with NRC-VAD. These results are consistent with the findings from the Weibo data set shown in Table 7, column (a), suggesting that the method of lexicon construction is less critical than the overall quality, relevance, and appropriateness of a lexicon (e.g., core versus peripheral) for the intended task.

We selected the following four state-of-the-art user-level baseline models and compared their performance with that of KETCH with ESI under the optimal setting:

- SISMO (Sawhney et al. 2021) is a hierarchical attention model optimized to factor in the graded nature of increasing suicide risk levels. It contains four components: a post embedding layer, a sequential context modeling layer, a temporal attention layer, and an ordinal regression layer. It first leverages Longformer, a pretrained transformer language model, to obtain post-level embeddings. Then, it introduces a BiLSTM layer to capture temporal features across multiple posts of a user.

- SDCNL (Haque et al. 2021) incorporates an unsupervised component to verify and correct noisy labels. We selected the Google Universal Sentence Encoder model for embedding and the k-means clustering algorithm for label denoising based on the original study. Additionally, we concatenated all posts of the same user.

- MentalRoBERTa (Ji et al. 2022a) is initialized with RoBERTa-based weights and trained with a large number of mental-health related posts collected from Reddit. We used a fully connected layer as the classifier, considered different model configurations, such as different aggregation methods and different numbers of historical posts, and reported the performance of the best model.

- XML-RoBERTa (Martínez-Castaño et al. 2020) is introduced in Section 4.3.2.

The results, as reported in Table 9, show that the KETCH-based model with the ESI lexicon significantly outperforms all the baseline models except the recall of SDCNL, as well as CLaC with a flagged F1 of 0.922 (Zirikly et al. 2019). We further segmented the Reddit users in the test set based on their total number of posts and examined the performance of KETCH for different user groups. The results presented in Table 10 do not exhibit

Table 9. Performance of KETCH-Based and Baseline User-Level Models on the Reddit Data

	Precision	Recall	F1	Accuracy
SDCNL	0.804	1.000	0.826	0.832
SISMO	0.902	0.774	0.816	0.738
MentalRoBERTa	0.869	0.815	0.835	0.872
XML-RoBERTa	0.775	0.757	0.762	0.806
KETCH with ESI	0.923	0.955	0.939	0.912

Notes. Values in bold represent the best performance results for each evaluation metric. We adopted the flagged F1 score (Zirikly et al. 2019), which is the F1 score for binary classification of flagged (suicide risk, including low, moderate, and high risks) vs. nonflagged (no suicide risk) classes.

a clear pattern in the relationship between the number of posts that a user has generated and model performance. Additionally, there are no significant differences in KETCH's performance among the different user groups. These results provide additional evidence for the robustness of KETCH.

5.4. Experiment 4: Generalizability of KETCH

We evaluate the generalizability of KETCH by applying it to the detection of depression, a different type of mental health problem, using the Reddit Self-Reported Depression Diagnosis (RSDD) data set (Yates et al. 2017). The RSDD data set contains 9,210 diagnosed users and 107,274 control users. On average, each user in the data set has 969 posts. The data set was relatively evenly divided into training, validation, and test sets based on random stratified sampling, each consisting of more than 39,000 posts.

We trained and tested a KETCH model and two baseline models for the detection of users with depression using the RSDD data set. Based on the results of experiment 3, we applied the averaging method to aggregate the five most recent posts of a user as model input. Cross-validation was done using TPOT. In addition, we used MentalRoBERTa, a pretrained language model for mental healthcare, for embedding learning and fine-tuned it with the training data. We created a depression lexicon of 252 terms by leveraging existing lexicons (Neuman et al. 2012, Choudhury et al. 2013, Cheng et al. 2016). We chose two baseline models: fine-tuned MentalRoBERTa (Ji et al. 2022a) and the CNN-based user model of Yates et al. (2017).

Table 10. Performance of the KETCH-Based Models for Users with Different Numbers of Posts

User group (user count)	Precision	Recall	F1	Accuracy
1-post users (94)	0.892	0.967	0.928	0.904
2-post users (17)	1.000	1.000	1.000	1.000
3-post users (6)	0.875	1.000	0.933	0.882
4-post users (2)	1.000	1.000	1.000	1.000
5 or above-post users (6)	1.000	1.000	1.000	1.000

Table 11. Performances of Depression Detection Models on the RSDD Data Set

Models	Precision	Recall	F1	Accuracy
Yates et al. (2017) model	0.59	0.45	0.51	Not available
Fine-tuned MentalRoBERTa	1.000	0.333	0.500	0.949
KETCH with MentalRoBERTa	0.736	0.806	0.765	0.924

The performances of different depression detection models are reported in Table 11. Their comparisons reveal that KETCH with the text representation learned by MentalRoBERTa achieved higher precision, recall, and F1 than the model of Yates et al. (2017) and much higher recall and F1 than the fine-tuned MentalRoBERTa model. The results demonstrate the good applicability and generalizability of KETCH to the detection of a different mental health problem.

5.5. Field Study for Feasibility Analysis

Proof of value is one of the key steps in design science research (Gill and Hevner 2013), presenting a pathway for a proposed model or system to achieve real impact. We conducted an online field study to explore the feasibility and user perceptions of applying KETCH to social media-enabled online intervention (KOI).

Existing research on social media-based intervention for people with SI has mainly focused on preventative intervention through training or education (e.g., providing guidelines or tips on how to manage stress) (Robinson et al. 2015, Perry et al. 2016). However, research on proactive and social media-based intervention in response to SI remains scarce. In our study, KOI first detected those posts involving SI by monitoring new posts on the Sina Weibo platform using KETCH, which automatically triggered alerts to a group of 16 volunteer consultants who were certified psychiatrists or psychologists. Each post would then be manually checked by

one of the consultants to confirm that its content indeed reflected SI. This verification step revealed an 86% detection accuracy. For each verified post, a consultant initiated contact with its creator through a private chat messaging service provided by Sina Weibo, inviting him/her to respond to a pre-intervention survey and schedule a free one-on-one virtual consultation via private chat on the same platform. The intervention procedure followed problem-solving therapy (D’zurilla and Goldfried 1971), a cognitive-behavioral intervention method. Finally, the participants were asked to respond to a postintervention survey about their perceptions of KOI. The questionnaire items were rated on a seven-point Likert scale, with 1 denoting strongly disagree, 4 neutral, and 7 strongly agree. Appendix D provides further procedure details.

In this field study, the consultants initiated contacts with 7,817 users whose posts were detected by KETCH as indicating SI. Of those users, 2,086 responded to the preintervention survey; 2,001 replied to the initial message from a consultant; and 100 participated in the one-on-one online intervention. Of the users who responded to the preintervention survey, 97.3% reported having some levels of depression; 76.7% had physically hurt themselves at least once in the past; 92.2% experienced medium or high levels of pain; 71.3% had medium or high levels of desire to die; and 94.1% had little or moderate hope for life. The majority of respondents (66.2%) had not received any prior psychological interventions, demonstrating the need for and benefits of social media-based detection and proactive intervention of suicidal ideation. About 70% of participants considered proactive contact (enabled by KETCH) from a consultant acceptable. The responses to the postintervention survey are summarized in Table 12 (see Appendix D).

The results of two-tailed one-sample *t* tests show that the participants’ acceptance of KETCH-enabled proactive contact, privacy protection, timely assistance,

Table 12. Descriptive Statistics of Postintervention Survey Responses

Perception of KOI	Mean	Standard deviation
Overall satisfaction	5.06	1.47
Acceptance of proactive contact	4.83	1.76
Privacy protection	5.37	1.51
Timely assistance	4.68	1.66
Continuous use intention	4.58	1.64
Recommendation to others	4.46	1.81
Comparison of KOI vs. face-to-face and phone-based interventions		
More willingness to engage	5.7	1.21
More personalized suggestions	4.9	1.52
Better understanding of my problem(s)	4.69	1.14
Easier to find a solution	4.69	1.25
More openness	5.13	1.02
Higher use intention	5.4	1.10

continuous use intention, and their overall satisfaction with KOI were all greater ($p < 0.001$) than neutral (i.e., 4 on the Likert scale). Most respondents indicated their willingness to recommend KOI to others who may experience SI ($p < 0.05$). Compared with face-to-face and phone-based interventions, the respondents were more willing to engage with and use KOI ($p < 0.001$), and felt that KOI provided more personalized suggestions, made them more open to the discussion of their problems, was better at understanding their problems, and helped them find a solution more easily ($p < 0.001$). These findings provide preliminary evidence for the practical value of KETCH in enabling social media-based intervention in the real-world setting.

6. Discussion

Suicide is a critical public health problem. SID has the potential to prevent serious and irreversible consequences. The openness, ubiquity, and anonymity of social media platforms encourage users to share their thoughts and emotions, including SI, creating opportunities for SID from social media.

6.1. Major Findings

In this study, we propose a transformer-based approach enhanced by a domain lexicon for SID from social media content. Our evaluation results show that integrating core lexicon-based domain knowledge can boost the performance of SID. In addition, our SI lexicon is more effective than an SI lexicon generated by ChatGPT, which proves the value of human expert involvement in domain lexicon development.

Our novel representation enhancement mechanism via model alignment is effective in improving SID, and both lexicon-term similarity and contextual importance have positive influences on model performance. KETCH consistently outperforms all baseline models across performance measures. Furthermore, this study demonstrates the robustness and generalizability of the proposed KETCH model. Beyond post-level models, we apply KETCH to build effective user-level models for suicide risk prediction and depression detection. The proposed approach to incorporating a domain lexicon into a transformer-based model is generic and applicable to other mental health issues. Although concatenation is the most commonly deployed aggregation method for developing user-level models for SID to date, our findings suggest that the averaging method generally outperforms concatenation and ensemble methods.

6.2. Research Contributions

This study follows the design science research paradigm and makes multifold contributions. We propose KETCH, a novel transformer-based approach enhanced

by domain knowledge. To the best of our knowledge, it is the first approach that incorporates a core social media-oriented domain lexicon into a state-of-the-art transformer model for SID. We find strong evidence that integrating domain-specific knowledge can boost the performance of SID. Moreover, KETCH goes beyond the lexicon-based enhancement by integrating it with aligned dynamic embedding built on representation refinement. This design enables KETCH to address both domain relevance and contextual importance in learning the representations of terms in social media posts. The proposed representation enhancement via model alignment improves SID performance.

From a knowledge integration perspective, this study makes the first attempt at designing a model-level method for integrating knowledge and embeddings. We propose a novel knowledge integration method that aligns text representation with knowledge units through the redesign of a transformer-based model. It offers a new methodological contribution to both NLP and SID communities, paving the way for integrating domain knowledge into a transformer-based model effectively to improve its performance.

From a model evaluation perspective, this research deploys a rigorous evaluation of the proposed IT artifacts that provides empirical insights on how to improve the rigor, relevance, and generalizability of design science research by using data collected from different social media platforms in different languages; by evaluating the generalizability of KETCH for depression detection; and by exploring the feasibility of KETCH for social media-enabled intervention through a field study. These evaluation methods offer new dimensions and insights for evaluating future models for SID from social media texts.

Through the aforementioned methodological contributions, our research is relevant to and contributes to the emerging information systems (IS) literature on artificial intelligence (AI) and healthcare analytics in multiple significant ways. First, recent years have witnessed rapid growth in deep learning-based NLP research across various topics in the IS community, such as financial risk prediction (Yang et al. 2023b), personality detection (Yang et al. 2023a), and customer misbehavior (Wu et al. 2021). However, there is a dearth of studies that redesign a model with human domain knowledge to optimize its performance. This study enhances a deep learning approach by infusing domain knowledge and dynamic embeddings, demonstrating the advantage of improving cutting-edge language models with domain knowledge to tackle a significant social challenge. Second, human-AI teaming research has recently received considerable attention (Fügener et al. 2021, Van den Broek et al. 2021). Most existing studies focus on how humans leverage AI for better decision making. Our study focuses on

effectively incorporating human domain knowledge into AI models, demonstrating the synergy between human intelligence and machine intelligence for performance improvement. Third, this study enriches existing IS research on leveraging machine learning techniques for social goods (Gerald et al. 2021, Xie et al. 2022), broadening the understanding of how technology can positively impact society. Fourth, this research extends the growing IS literature on health IT and online health communities (Ayabakan et al. 2017, Lin et al. 2017, Chau et al. 2020, Liu et al. 2020, Saifee et al. 2020, Zhang and Ram 2020) by proposing a domain knowledge-enhanced solution to SID. Last but not least, diverging from the IS norm of single-source data reliance (Etudo and Yoon 2023), our multilingual and multilevel (i.e., post-level and user-level) experiment settings, which include social media posts in Chinese and English and involve different problems (i.e., SID and depression detection), underscores KETCH's profound research implications.

6.3. Practical Implications

Our proposed KETCH and study findings offer practical implications for various stakeholders. KETCH can detect social media users with SI, who may otherwise go unnoticed, by monitoring their social media posts and making timely intervention possible. Most people with mental health issues do not visit healthcare professionals for help because mental illness remains heavily stigmatized. Therefore, their engagement with others on social media platforms holds great promise for the detection and intervention of SI. For example, college students have experienced increased mental pressure and stress. Suicide rates on campuses have been on the rise: 6% of undergraduates and 4% of graduate students have seriously considered attempting suicide (Adams et al. 2017). According to the U.S. Centers for Disease Control and Prevention,⁷ in 2021, suicide was the second leading cause of death for people ages 10–14 and 20–34 years. Given that college students and teenagers are major users of social media, KETCH provides a valuable tool for identifying students with SI on social media platforms and reducing the number of lives lost to suicide.

Our work provides mental health professionals and organizations with an alternative means to the traditional in-person methods for suicide risk assessment by identifying and enabling intervention for individuals who experience SI but do not actively seek professional assistance. Given the increasing shortage of mental health professionals, even people with SI who want to seek help may be unable to access a mental health expert promptly. KETCH has the potential to alleviate this problem by leveraging social media data and the wide reach and availability of the delivery platform. In addition, KETCH can help ease the workload of mental

health professionals in identifying potential patients. Appendix E presents an analysis of the potential economic benefits of KETCH, estimating that KETCH may be able to help prevent approximately 1,800 suicide deaths in the United States annually, which leads to \$18.9 billion in cost savings. More in-depth future research is needed to accurately measure the economic impact of such SID models and enable intervention.

For a social media platform, leaving user posts involving SI unattended can be perceived as the platform failing to protect its users and the public good, and this may harm the platform's reputation or even expose it to legal liabilities. KOI provides a preliminary guide for how to design and deploy an end-to-end solution that seamlessly integrates automatic SID and intervention. Social media platforms can deploy KETCH to enable timely intervention, such as sharing online mental health management resources, offering psychiatric consulting services, and providing contact information of mental health organizations. Furthermore, our findings have implications for policymakers responsible for formulating and amending policies and regulations regarding the corporate responsibilities of social media platforms for public welfare.

To maximize the practical value of KOI, the right balance between recall and precision needs to be carefully calibrated given the intrinsic tradeoff between the two (Davis and Goadrich 2006). SID from social media content is a delicate task with high stakes. On one hand, prioritizing recall ensures the detection of as many SI posts as possible for timely intervention, which is crucial for saving lives, improving well-being, increasing productivity, and reducing societal costs. On the other hand, higher recall may lead to more false positives and lower precision. False positives can result in substantial costs by misleading mental health professionals to waste their time on trying to assist social media users who actually do not have SI and by diverting healthcare resources and services away from individuals who can truly benefit from KOI designed for early intervention and prevention of suicide. This is particularly critical given the severe shortage of mental health professionals on a global scale. As of December 2023, more than half of the U.S. population lives in a mental health professional shortage area.⁸ In addition, false positives can confuse or frustrate individuals by subjecting them to unwarranted contacts and interventions, resulting in unnecessary distress, anxiety, and healthcare costs. Therefore, practitioners should carefully assess both costs and benefits, as exemplified in the approach of Weiner et al. (2018), to determine a socially optimal level of intervention.

Although we propose KETCH in the context of SID and investigate its generalizability in depression detection, the overall methodological design of KETCH, particularly

the proposed approach to integrating a domain lexicon into a transformer-based model, applies to other NLP tasks that can benefit from domain knowledge. For example, it is reasonable to assume that KETCH, with different domain lexicons, is capable of detecting other mental disorders or offensive content from social media posts.

6.4. Limitations and Future Research

This research can be extended in multiple directions in future studies. First, like most existing studies, KETCH focuses only on the textual content of social media posts. Future research can leverage multimodal representations of social media posts (e.g., both text and image) to enhance the effectiveness of detection models. Second, a social media-based lexicon, such as our Chinese SI lexicon, may require periodic updates, particularly when there are significant changes in the terms and expressions used in social media posts with SI, to maintain the SID model's effectiveness. Such updates are likely to be incremental, such as adding new relevant and important terms, rather than removing existing lexicon terms or rebuilding a new lexicon from scratch. Thus, the associated cost of these lexicon updates should be manageable. Additionally, both the quantity and quality of terms within a core lexicon can influence the performance of KETCH. Consequently, it is advisable for future research to develop design guidelines pertaining to these factors. Third, the primary objective of the field study on KOI is to explore its feasibility rather than re-evaluate the detection performance of KETCH. A rigorous evaluation of KOI is a complex and separate research problem that is beyond the scope of this study, which requires future research with more sophisticated design and control. Fourth, our data sets have some limitations. Although our Sina Weibo data set retains the sequence of postings, it lacks specific post timestamps. As a result, we were unable

to analyze the effect of elapsed time on the onset of SI. Furthermore, considering user privacy and our research aims, we did not collect user demographic information in the field study. Through directly collecting or indirectly inferring user demographic information from social media sites or other data sources, future studies can examine the robustness of KETCH with subgroups of users with different demographic characteristics in postmodel analyses. By combining such an approach with timestamps, future research can also gain insights into how the mental state of different groups of users with SI may change over time and predict the urgency of intervention. Finally, human biases and inconsistencies may inadvertently occur in the expert-curated lexicon construction process. Humans are susceptible to individual biases due to different experiences, knowledge, personal preferences, and cognitive processes (Johnson 2021). Consequently, using different expert panels may result in some variations in the lexicon because different experts may identify different seed terms or judge the relevancy of individual terms to SI in slightly different ways. In this study, we employed multiple measures to mitigate such biases during lexicon construction, including providing specialized training to experts, involving at least three experts in each human-intervention step, and implementing interrater reliability and majority voting. Future research should explore formal methods for mitigating human biases and variations in lexicon construction.

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Appendix A. Chinese SI Lexicon

一个人走	厌倦	想死	死	破灭
一了百了	去死	想自杀	死一死	离别
一切都不再重要	受不了	想要死掉	死了算	离去
一命归西	受够	想解脱	死亡	离开
一死	另一个世界	想逃走	死去活来	算了
一生	只能这样了	我不在也没差	死后	累了
一走了之	同归于尽	我去死	死掉	终了
一起死	后会无期	我已经不在	死期	终点
一辈子	含笑而逝	我死后	死神	终结
上苍	告别	我的离开	死给你看	结局
下场	告别会	我走了	死路	结束
下辈子	告别式	投胎	殉葬	绝命书
不会再见	告别父母	报应	残酷的命运	绝境
不再	告终	抱歉	毁灭	绝笔
了无生趣	噩梦	拜拜	永不再会	绝路
不可能	回头路	挂了该多好	永不再见	继承人
不在了	因果	挂念	永别	缘尽
不如归去	地狱	挂牵	永恒	老天
不孝	坚持不下去	挽救自己	永诀	耗尽
不得不走	埋葬	挽联	永远	脱离苦海
不得已	堕入地狱	提前离开	永远不见	自寻短见
不必在意	多余	摆脱	求死不得	自尽
不必想我	大逆不道	撇下	没了我	自我了断
不想活	天国	撑不下去	没什么好怕	自戕
不想继续活下去	天堂	撒手不管	没关系	自杀
不活	天要亡我	撒手闭眼	没意思	舍不得
不用想我	天谴	操蛋的人生	没救了	舍弃
不能承受	太累	放弃	没有以后	苟活
不舍	失去我	救救我	没有价值	落入地狱
不要想我	孤苦伶仃	整理遗物	没有再见	落幕
不要活	完了	无情的世界	没有意义	要死要活
不适应这个世界	完结	无意义	没有明天	解脱
丢下	完蛋	无憾	没有机会	记挂
丧生	对不住	无所留恋	没有用	诀别
为时已晚	对不起	无所谓	没法活	该死
举殡	对不起先走	无望	没活路了	负担
了却	对不起爸妈	无法	活	走投无路
了断	寻了短见	无法活	活下去	转世
了结	寻死	无缘	活不下去	轮回
亡	尘世	无能为力	活得好累	轻生
人生	尽头	无话可说	活着不容易	运数
人生不过如此	尾声	无间地狱	活着为什么	这个世界
今世	已尽力	日子没法过	活得很累	选择退出
今生	已经不在	时间到了	活路	逼上绝路
今生无缘	已经死去	托生	消失	逼我死
代价	已经结束	是时候	消极	逼死
任性	带我走	瞋目	消逝	道别
企图自杀	彻底	最后	深渊	遗书
会死的	彻底结束	最后一天	濒临	遗嘱
依恋	往生	最后一次	炼狱	遗愿
倒计时	很累	最后一站	牵挂	遗物
偿命	心如古井	最后告别	现世报	遗言
先走一步	心如死灰	最后的挣扎	生不逢时	遗骨
再也	心已死	未来	生不如死	都没了
再见	心死	杀	生命	都要好好的
决绝	心累	杀了我吧	生无可恋	阿鼻地狱
别了	忍够了	杀了自己	生死	陪葬
别无出路	怎么办	来不及	生路	难以为继
到头	怎么活	来世	留不住	静静地
到此为止	惊醒	来生	留恋	非死不可
到终点	惦记	极乐世界	疲惫	面对死亡
割舍	想不开	棺材	疲累	骨灰
半死	想去死	此生	痛不欲生	黄泉
厌世	想摆脱	步入地狱	看淡	黑暗

Appendix B. Sample Social Media Posts

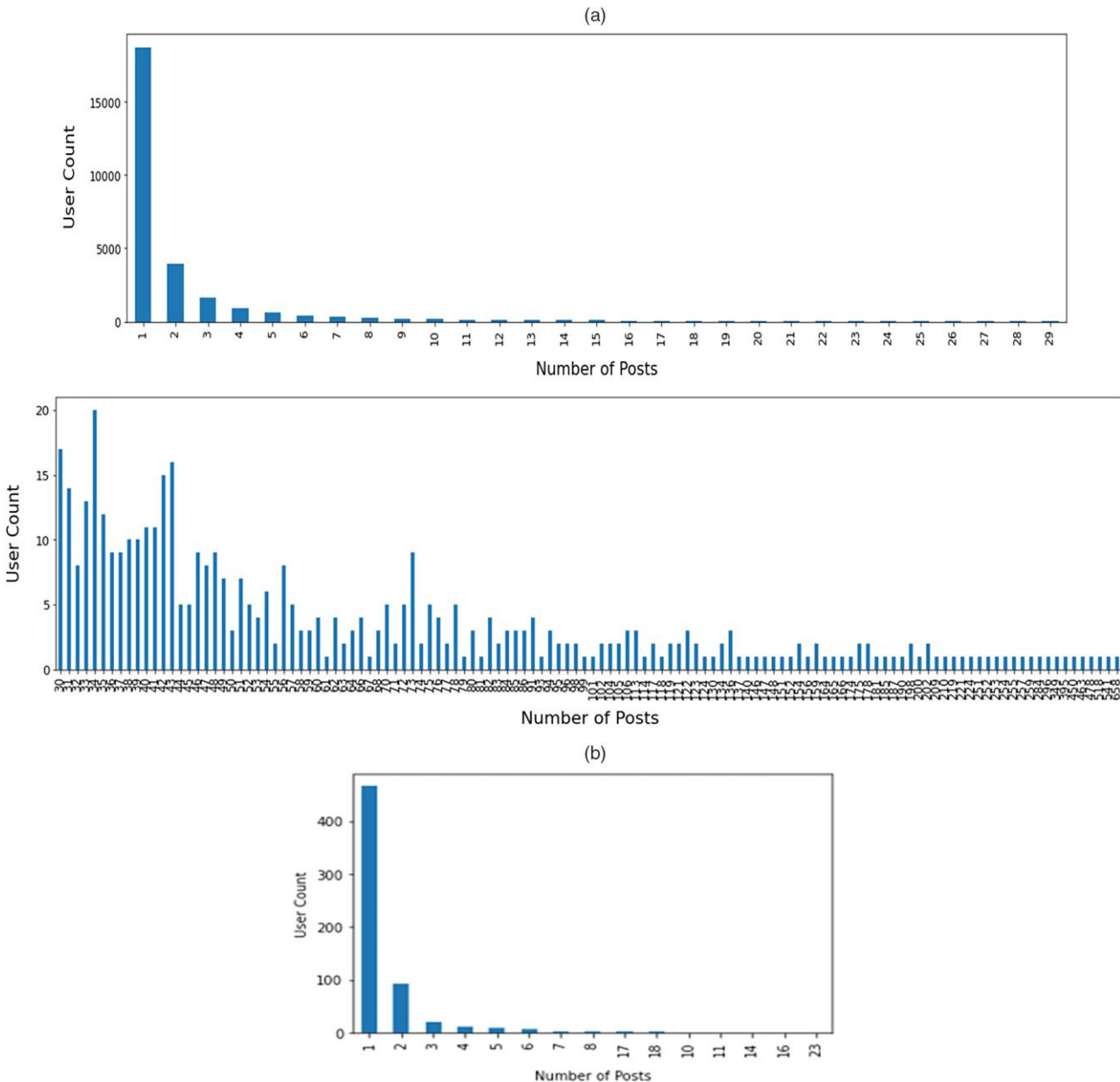
SI posts with SI terms	关注你三年了。很少会留言。很想抛弃我身上的所有枷锁，一死了之。 (I have been paying attention to you for three years, yet have rarely left any messages. Want to free myself from all the shackles locked on me and end my life.) “我讨厌我自己，讨厌到恨不得杀了自己。” (I am so sick of myself that I want to kill myself .) 我想离开这个让人活得好累的世界。 (I want to leave this world that makes me feel so tired.)
Non-SI posts with SI terms	算了，反正也没胃口。(Forget it. I don't have an appetite anyway) 能忍受死去的疼痛，为什么不能勇敢的活下去? (If you could bear the pain of dying , why not have the courage to live in this world?)
SI posts without SI terms	生活真艰难。我真的挺不了太长时间了。(Life is so hard. I really cannot stand it for much longer.) 有一天，我也会走上你的路。(One day, I will walk the same path as you.)

Appendix C. Descriptive Statistics of the Chinese Weibo Data set

Figure C.1 shows the distributions of user counts across the varying number of posts. The descriptive statistics of the number of

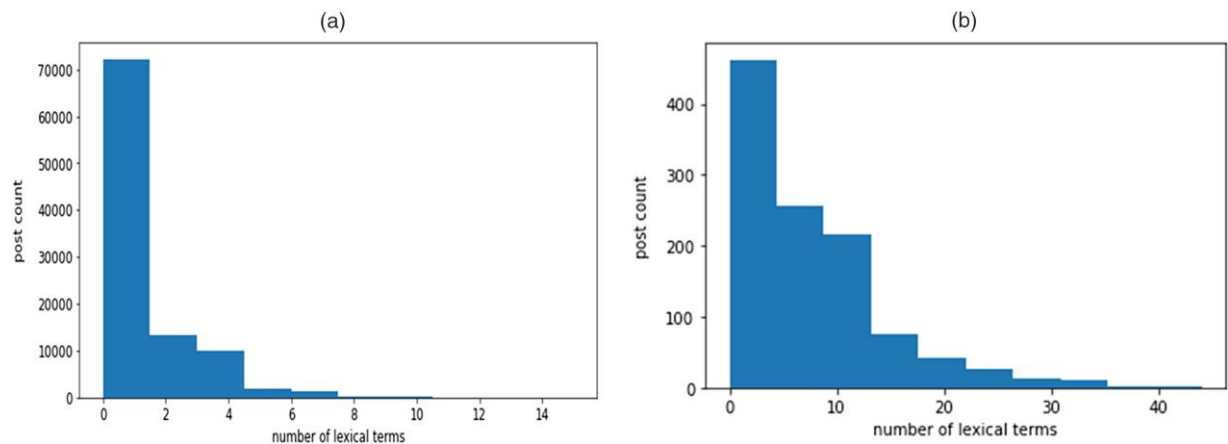
posts per user are as follows: Chinese Weibo data (mean = 3.47, standard deviation = 13.55) and English Reddit SI data (mean = 1.66, standard deviation = 2.09). Figure C.2 shows the distributions of posts across the varying number of lexicon terms.

Figure C.1. (Color online) Distribution of Users Across Varying Number of Posts



Notes. Users with fewer than 30 posts. Users with 30 or more posts. (a) Chinese Weibo data. (b) English Reddit data.

Figure C.2. (Color online) Frequency Distributions of Posts Containing Lexicon Terms



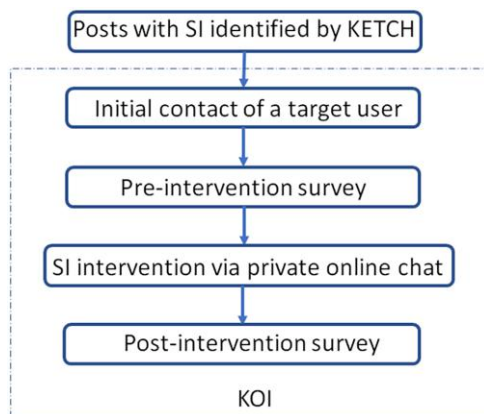
Notes. (a) Chinese Weibo data. (b) English Reddit data.

Appendix D. Field Study

D.1. Intervention Procedure

Figure D.1 presents the process flow of the field study for a comprehensive end-to-end solution that includes both continuous detection of and intervention for SI on social media. The solution consists of two main modules: one is KETCH, and the other is KETCH-enabled online intervention (KOI). In our online field test, the solution was prototyped in the following way: after KETCH detected suicidal ideation in a Sina Weibo user's post, a consulting psychologist would be notified and initiate contact with the target user, who was a stranger, by sending a private message through the Sina Weibo platform. In the initial private message, the consultant disclosed that he/she had read the target user's most recent post and noticed the latter's low mood; expressed concern for the user and willingness to help; briefly introduced the voluntary and free psychological consulting service; provided a URL leading to a pre-intervention psychological assessment survey; and offered tips for stress management and provided a mental health emergency center hotline number. For individuals who were found to have relapses of suicidal ideation, we prepared several different versions of initial text messages that would be used in a certain sequence.

Figure D.1. (Color online) Process Flow of the Field Study



In response to the initial message from the consulting psychologist or psychiatrist, the target users could choose to take one or more of the following actions:

- (1) Visiting the website and responding to the online pre-intervention survey. The survey consisted of demographic questions, self-assessment of perceived pain, desire to die, and hope for life on a scale of 0-100, the PHQ-9 questionnaire for psychological assessment, and previous experience with suicidal ideation and seeking intervention.
- (2) Scheduling an intervention session by responding to the consultant's private message directly.
- (3) Contacting and receiving assistance from the mental health emergency center using the contact information provided in the initial message.
- (4) Doing nothing.

If a target chose to schedule an intervention session, the consultant would re-contact the target to perform the standard personalized PST-based intervention procedure, (D'zurilla and Goldfried 1971) commonly used in medical practice, through the private online chat on Sina Weibo. Upon completing the intervention, the target individual would be asked to answer a post-intervention questionnaire to assess his/her perception and acceptance of the intervention. Based on our experience, individuals with suicidal ideation are often reluctant to respond to a survey, let alone a lengthy one. To increase the response rate of this special group of users, we developed a short form of the post-intervention survey. The survey items were measured on a seven-point Likert scale, with 1 indicating "strongly disagree," 4 "neutral," and 7 "strongly agree."

D.2. Postintervention Assessment

Please answer the following questions using the following scale: 1: Absolutely disagree; 2: Disagree; 3: Somewhat disagree; 4: Neither agree nor disagree; 5: Somewhat agree; 6: Agree; 7: Absolutely agree

Part 1: Based on the overall communication and consulting experience that I had with the consultant through the private chat on Sina Weibo,

1. Overall, I am satisfied with this private-chat based online consulting.

2. I am comfortable with you proactively reaching out to me through private text messages.
 3. The private-chat based online consulting protects my privacy.
 4. The private-chat based online consulting assisted me in a timely manner.
 5. I would like to use this private-chat based online consulting again in the future when I need help.
 6. I would like to recommend this private-chat based online consulting to others who may need help.
- Part 2: In comparison with face-to-face and telephone-based psychological consulting,
7. The private-chat based online consulting makes me more willing to engage in communication with the consulting psychologist to find a solution to the problem(s) I am facing.
 8. The private-chat based online consulting provides me with more personalized suggestions based on my actual situation.
 9. The private-chat based online consulting that is problem-solving oriented helps me better understand my problem(s).
 10. The private-chat based online consulting that is problem-solving oriented helps me find a solution to my problem(s).
 11. I feel that I can discuss more problems with the consulting psychologist.
 12. I am more willing to consult through such an online private chat.

Appendix E. Economic Analysis

To estimate the potential economic value of KETCH, we performed the following economic analysis. According to the CDC, in 2019, suicide and nonfatal self-harm cost the United States nearly \$490 billion, which includes medical costs, work loss costs, and quality of life costs.⁹ The CDC reported 47,511 suicide deaths in the United States in 2019.¹⁰ This is consistent with the report from Conner et al. (2019) that the suicide death rate is 15 per 100,000 person-years for individuals 15 years or older, and the suicide attempt rate is 175 per 100,000 person-years. We therefore use these ratios as the basis of our analysis. These numbers were also adopted by Ross et al. (2021) for cost-effective suicide risk prediction.

According to Statista, the United States had 278 million active social media users in 2019.¹¹ Applying the above suicide attempt rate and suicide rate, we estimate that there would be 486,553 suicide attempts and 41,705 suicide deaths each year. We assume that 20% of suicide attempts would be reflected in and detected from social media content. Given the 88.1% recall and 25% of intervention uptake rates (Guthrie et al. 2001), KETCH will be able to help prevent 1,837 suicide deaths, which leads to \$18.9 billion in cost savings annually. Moreover, should suicidal ideation remain undetected or not detected in time, it could result in suicide attempts and reoccurring treatments, which significantly increases the national healthcare cost.

The GDP per capita of the United States in 2021 was \$69,288. Considering that youth and young adults aged 10 to 24 account for 14% of all suicide incidents, the economic benefit of suicide detection and prevention is significant. For example, a 24-year-old with average economic productivity who does not commit suicide would go on to produce millions of dollars of value for the economy if the retirement age

is estimated to be 65 and the average GDP growth rate is 2.5%.

The previous estimate of the economic value of KETCH is conservative for two reasons. First, the CDC estimate is largely based on the Web-based Injury Statistics Query and Reporting System (WISQARS), which may severely underestimate the magnitude of suicide attempts compared with the actual cases (Shepard et al. 2016). Second, younger generations tend to be more active on social media, and their suicidal ideation therefore will be more likely to be captured by KETCH. Because younger age is associated with higher values of quality of life (Peterson et al. 2021), the benefit and cost savings of preventing suicide in younger groups can be even greater.

Endnotes

- ¹ Adapted from <https://www.theringer.com/2017/4/24/16038130/social-media-suicide-prevention-policies-5490c2c224e0>, Accessed on May 4, 2021.
- ² See <https://www.who.int/news-room/fact-sheets/detail/suicide>, Accessed on December 1, 2023.
- ³ See <http://www.nami.org/mhstats>, Accessed on December 14, 2023.
- ⁴ See <https://www.nimh.nih.gov/health/topics/depression>, Accessed on December 23, 2023.
- ⁵ For example, since 2016, Facebook users have been allowed to flag questionable or concerning posts for the company to review. Its community operations team then reviews the flagged posts to decide if Facebook should provide crisis resources to individuals.
- ⁶ See <https://weibo.com/xiaofan116>, Accessed on March 14, 2019.
- ⁷ Web-based Injury Statistics Query and Reporting System (WISQARS) [online]. [Feb 17, 2022]. Available from: www.cdc.gov/injury/wisqars.
- ⁸ See bhw.hrsa.gov/sites/default/files/bureau-health-workforce/Behavioral-Health-Workforce-Brief-2023.pdf, Accessed on January 5, 2024.
- ⁹ See <https://www.cdc.gov/suicide/facts/index.html>, Accessed on September 17, 2023.
- ¹⁰ See https://www.cdc.gov/mmwr/volumes/71/wr/mm7108a5.htm#T1_down, Accessed on September 17, 2023.
- ¹¹ See <https://www.statista.com/statistics/278409/number-of-social-network-users-in-the-united-states/>, Accessed on February 2, 2024.

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